

Hybrid_Pruning

December 14, 2025

```
[2]: import argparse
import os
import time
import shutil

import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
import torch.backends.cudnn as cudnn

from tensorboardX import SummaryWriter

import torchvision
import torchvision.transforms as transforms

import torch.nn.utils.prune as prune

from models import *

global best_prec
use_gpu = torch.cuda.is_available()
print('=> Building model...')

batch_size = 128
model_name = "VGG16_quant"
model = VGG16_quant()
print(model)

normalize = transforms.Normalize(mean=[0.491, 0.482, 0.447], std=[0.247, 0.243, 0.262])

train_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=True,
```

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download=True,
transform=transforms.Compose([
    transforms.RandomCrop(32, padding=4),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    normalize,
]))
trainloader = torch.utils.data.DataLoader(train_dataset, batch_size=batch_size,
↳shuffle=True, num_workers=2)

test_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=False,
    download=True,
    transform=transforms.Compose([
        transforms.ToTensor(),
        normalize,
    ]))

testloader = torch.utils.data.DataLoader(test_dataset, batch_size=batch_size,
↳shuffle=False, num_workers=2)

print_freq = 100 # every 100 batches, accuracy printed. Here, each batch
↳includes "batch_size" data points
# CIFAR10 has 50,000 training data, and 10,000 validation data.

def train(trainloader, model, criterion, optimizer, epoch):
    batch_time = AverageMeter()
    data_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    model.train()

    end = time.time()
    for i, (input, target) in enumerate(trainloader):
        # measure data loading time
        data_time.update(time.time() - end)

        input, target = input.cuda(), target.cuda()

        # compute output
        output = model(input)
        loss = criterion(output, target)

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    # measure accuracy and record loss
    prec = accuracy(output, target)[0]
    losses.update(loss.item(), input.size(0))
    top1.update(prec.item(), input.size(0))

    # compute gradient and do SGD step
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    # measure elapsed time
    batch_time.update(time.time() - end)
    end = time.time()

    if i % print_freq == 0:
        print('Epoch: [{0}] [{1}/{2}]\t'
              'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
              'Data {data_time.val:.3f} ({data_time.avg:.3f})\t'
              'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
              'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                epoch, i, len(trainloader), batch_time=batch_time,
                data_time=data_time, loss=losses, top1=top1))

def validate(val_loader, model, criterion ):
    batch_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    # switch to evaluate mode
    model.eval()

    end = time.time()
    with torch.no_grad():
        for i, (input, target) in enumerate(val_loader):

            input, target = input.cuda(), target.cuda()

            # compute output
            output = model(input)
            loss = criterion(output, target)

            # measure accuracy and record loss
            prec = accuracy(output, target)[0]
            losses.update(loss.item(), input.size(0))

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        top1.update(prec.item(), input.size(0))

        # measure elapsed time
        batch_time.update(time.time() - end)
        end = time.time()

        if i % print_freq == 0: # This line shows how frequently print out
            ↪ the status. e.g., i%5 => every 5 batch, prints out
            print('Test: [{0}/{1}]\t'
                  'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
                  'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
                  'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                      i, len(val_loader), batch_time=batch_time, loss=losses,
                      top1=top1))

    print(' * Prec {top1.avg:.3f}% '.format(top1=top1))
    return top1.avg

def accuracy(output, target, topk=(1,)):
    """Computes the precision@k for the specified values of k"""
    maxk = max(topk)
    batch_size = target.size(0)

    _, pred = output.topk(maxk, 1, True, True)
    pred = pred.t()
    correct = pred.eq(target.view(1, -1).expand_as(pred))

    res = []
    for k in topk:
        correct_k = correct[:k].view(-1).float().sum(0)
        res.append(correct_k.mul_(100.0 / batch_size))
    return res

class AverageMeter(object):
    """Computes and stores the average and current value"""
    def __init__(self):
        self.reset()

    def reset(self):
        self.val = 0
        self.avg = 0
        self.sum = 0
        self.count = 0

    def update(self, val, n=1):

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        self.val = val
        self.sum += val * n
        self.count += n
        self.avg = self.sum / self.count

def save_checkpoint(state, is_best, fdir):
    filepath = os.path.join(fdir, 'checkpoint.pth')
    torch.save(state, filepath)
    if is_best:
        shutil.copyfile(filepath, os.path.join(fdir, 'model_best.pth.tar'))

def adjust_learning_rate(optimizer, epoch):
    """For resnet, the lr starts from 0.1, and is divided by 10 at 80 and 120_
    ↪ epochs"""
    adjust_list = [150, 225]
    if epoch in adjust_list:
        for param_group in optimizer.param_groups:
            param_group['lr'] = param_group['lr'] * 0.1

#model = nn.DataParallel(model).cuda()
#all_params = checkpoint['state_dict']
#model.load_state_dict(all_params, strict=False)
#criterion = nn.CrossEntropyLoss().cuda()
#validate(testloader, model, criterion)

```

=> Building model...

```

VGG_quant(
  (features): Sequential(
    (0): QuantConv2d(
      3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): QuantConv2d(
      64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (5): ReLU(inplace=True)
    (6): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (7): QuantConv2d(

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        64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (8): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (9): ReLU(inplace=True)
    (10): QuantConv2d(
        128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (11): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (12): ReLU(inplace=True)
    (13): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (14): QuantConv2d(
        128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (15): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (16): ReLU(inplace=True)
    (17): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (18): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (19): ReLU(inplace=True)
    (20): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (21): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (22): ReLU(inplace=True)
    (23): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (24): QuantConv2d(
        256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (25): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (26): ReLU(inplace=True)
    (27): QuantConv2d(
        512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()

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    )
    (28): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (29): ReLU(inplace=True)
    (30): QuantConv2d(
        512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (31): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (32): ReLU(inplace=True)
    (33): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (34): QuantConv2d(
        512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (35): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (36): ReLU(inplace=True)
    (37): QuantConv2d(
        512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (38): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (39): ReLU(inplace=True)
    (40): QuantConv2d(
        512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (41): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (42): ReLU(inplace=True)
    (43): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (44): AvgPool2d(kernel_size=1, stride=1, padding=0)
    )
    (classifier): Linear(in_features=512, out_features=10, bias=True)
    )

```

Downloading <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz> to
./data/cifar-10-python.tar.gz

100%| | 170498071/170498071 [00:05<00:00, 29768105.28it/s]

Extracting ./data/cifar-10-python.tar.gz to ./data
Files already downloaded and verified

```
[3]: # HW

# 1. Load your saved model and validate
# 2. Replace your model's all the Conv's weight with quantized weight
# 3. Apply reasonable alpha
# 4. Then, try to multiple bit precisions and draw graph of bit precision vs.
    ↳ accuracy
```

```
[4]: PATH = "result/VGG16_quant/model_best.pth.tar"
checkpoint = torch.load(PATH)
model.load_state_dict(checkpoint['state_dict'])
device = torch.device("cuda")

model.cuda()
model.eval()

test_loss = 0
correct = 0

with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))
```

Test set: Accuracy: 9273/10000 (93%)

```
[4]: ##### Prune all the QuantConv2D layers' 80% weights with 1) unstructured, and 2)
    ↳ structured manner.
```

```
[5]: i = 0
for layer in model.modules():
    i = i+1
    if isinstance(layer, QuantConv2d):
        prune.l1_unstructured(layer, name="weight", amount=0.8)
```

```
[6]: print(list(model.features[40].named_parameters())) # check whether there is
    ↳ mask, weight_org, ...
```



```
print(model.features[40].weight) # check whether there are many zeros
```

```
[('act_alpha', Parameter containing:
tensor(0.6882, device='cuda:0', requires_grad=True)), ('weight_q', Parameter
containing:
tensor([[[[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

          ...,

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.5025, -0.0000],
          [ 0.0000,  0.5025, -0.0000]],

          [[-0.5025, -0.5025, -0.0000],
          [-0.0000, -0.5025, -0.0000],
          [ 0.0000, -0.0000, -0.0000]],

          [[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.5025]],

          ...,

          [[-0.5025, -0.5025, -0.0000],
```

```

[ 1.0049, -0.5025, -0.5025],
[-1.0049,  1.5074, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
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 [-1.0049, -0.0000, -0.0000]],

[[-0.0000, -0.5025, -0.0000],
 [-0.5025, -0.5025, -0.0000],
 [ 0.0000, -0.0000, -0.0000]]],

[[[ 1.0049,  1.0049,  0.0000],
 [ 2.5123,  3.0147, -0.0000],
 [-0.0000,  0.5025, -0.0000]],

[[ 1.0049,  0.5025, -0.5025],
 [ 0.5025,  0.5025, -0.0000],
 [ 0.0000, -0.0000, -0.0000]],

[[-1.0049, -0.0000, -0.5025],
 [ 0.0000,  0.0000, -0.5025],
 [-0.0000, -0.5025, -0.0000]],

...,

[[-2.0098, -2.0098, -1.0049],
 [-1.5074, -1.0049, -1.0049],
 [-0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000,  0.0000],
 [-1.0049,  2.5123,  0.0000],
 [-0.0000,  0.5025,  0.0000]],

[[ 0.5025,  0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
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...,

[[[-0.0000, -0.0000, -0.0000],
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```

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[-0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
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 [-0.0000, -0.0000, -0.0000]],

...,

[[-0.0000, -0.0000, -0.0000],
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 [ 0.5025,  0.5025, -0.0000]],

[[ 0.0000, -0.0000, -0.0000],
 [ 1.0049, -0.5025, -0.5025],
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...,

[[-0.5025, -0.0000, -0.0000],
 [ 1.0049, -0.5025, -0.5025],
 [-0.5025,  0.5025, -0.0000]],

[[-0.5025, -0.0000, -0.0000],
 [ 0.5025, -0.0000, -0.0000],
 [-1.0049, -0.5025, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.5025, -0.0000, -0.0000],
 [ 0.0000, -0.0000, -0.0000]]],

```

```

[[[-0.0000, -1.0049, -0.5025],
  [ 0.5025,  2.0098, -1.0049],
  [ 3.5172,  0.0000, -0.0000]],

 [[-0.0000,  0.5025, -0.0000],
  [-0.5025,  3.0147, -1.5074],
  [ 2.0098,  3.5172,  0.0000]],

 [[-0.0000,  0.5025,  0.0000],
  [-0.5025, -1.0049, -0.0000],
  [-0.5025,  3.5172,  0.0000]],

 ...,

 [[-0.5025,  0.0000, -0.0000],
  [-3.5172, -3.5172, -1.0049],
  [ 3.5172,  3.5172, -0.0000]],

 [[-1.0049, -0.5025,  0.0000],
  [-2.5123, -2.5123, -0.5025],
  [ 3.5172,  3.5172, -1.0049]],

 [[ 0.5025,  0.5025,  0.0000],
  [ 1.0049,  0.5025, -0.0000],
  [ 1.0049, -0.0000, -0.0000]]]], device='cuda:0', requires_grad=True)),
('weight_orig', Parameter containing:
tensor([[[[-1.7570e-05, -8.9569e-06,  1.0557e-06],
  [ 1.6296e-05,  8.2165e-05, -6.3749e-06],
  [ 6.6472e-05,  4.8311e-05, -2.4037e-06]],

 [[-2.1359e-05,  4.6951e-05,  9.5514e-06],
  [-3.1503e-05,  2.1955e-05,  4.0229e-05],
  [ 3.5774e-05, -4.9366e-06, -1.1941e-05]],

 [[ 9.7124e-06, -4.1532e-06, -1.4637e-05],
  [ 3.3825e-05, -4.3020e-05,  7.5637e-05],
  [ 3.3258e-05,  1.9391e-05,  2.0105e-05]],

 ...,

 [[ 2.0706e-06, -4.3704e-06, -2.2272e-06],
  [ 1.1490e-05,  6.5538e-05, -1.2129e-05],
  [-2.7008e-05, -3.6061e-05, -2.4124e-05]],

 [[ 3.5951e-05, -3.2756e-05,  9.0803e-06],
  [ 7.9857e-05,  3.4086e-05, -1.6635e-06],
  [ 9.4075e-06,  1.1999e-04, -4.0382e-05]]],

```

[[1.8758e-05, 2.7689e-05, 1.7139e-05],
[-4.1422e-05, 9.5847e-08, 8.1808e-06],
[5.8145e-05, -1.9476e-05, -3.8287e-05]]],

[[[-1.5517e-04, 8.2275e-05, -1.4624e-04],
[-7.3287e-04, -8.5354e-04, -6.3418e-04],
[4.1919e-04, 2.4063e-03, -7.7785e-04]]],

[[[-1.2000e-03, -1.5387e-03, -1.7177e-04],
[-7.3789e-04, -1.6335e-03, -6.7902e-04],
[7.0406e-04, 2.3091e-05, -6.2861e-04]]],

[[[-1.2771e-04, -7.0946e-04, -1.3043e-04],
[-1.4423e-04, -1.2029e-04, -7.1849e-04],
[-8.7420e-05, -6.2121e-04, -9.8891e-04]]],

...,

[[[-1.2337e-03, -2.2048e-03, -3.9058e-04],
[4.8492e-03, -2.3796e-03, -9.1923e-04],
[-3.3027e-03, 5.4508e-03, -3.8683e-04]]],

[[[-7.4445e-05, -1.2986e-04, -6.3556e-05],
[4.9579e-03, 1.7841e-04, -4.9176e-05],
[-3.3219e-03, -2.3107e-05, -6.8390e-04]]],

[[[-7.4685e-04, -8.9520e-04, -7.8436e-05],
[-1.7259e-03, -1.4290e-03, -4.8922e-04],
[9.8038e-04, -4.4916e-05, -1.4341e-05]]],

[[[3.4294e-03, 3.8984e-03, 9.9814e-04],
[9.7354e-03, 1.1741e-02, -3.5083e-04],
[2.0531e-05, 2.5215e-03, 1.4772e-05]]],

[[4.3132e-03, 1.9929e-03, -1.8510e-03],
[1.7087e-03, 1.3521e-03, -1.9382e-04],
[1.8923e-04, -3.1905e-04, 5.1719e-05]]],

[[[-3.2476e-03, -1.4931e-04, -1.8248e-03],
[3.4393e-04, 7.2529e-04, -1.3105e-03],
[-6.9917e-06, -1.0477e-03, -1.8952e-04]]],

...,

[[[-7.8411e-03, -8.1294e-03, -2.9235e-03],
[-5.6326e-03, -4.1152e-03, -3.2639e-03],

```

[-5.9923e-04, -2.5031e-04, -2.4815e-04]],

[[-4.0059e-04, 7.0557e-05, 8.7646e-04],
 [-3.5183e-03, 1.0080e-02, 7.0322e-04],
 [-4.8108e-04, 1.4617e-03, 5.6965e-04]],

[[ 2.8139e-03, 1.0220e-03, -1.0918e-04],
 [-6.5041e-04, -3.5622e-04, 4.9510e-05],
 [ 3.6596e-04, 1.3369e-04, 4.9963e-05]]],

...,

[[[-3.3910e-05, -5.3211e-05, -9.4828e-05],
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 [-1.1428e-04, 1.5189e-04, -8.7737e-05]],

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grad_fn=<MulBackward0>)

```

```

[7]: ### Check sparsity ###
mask1 = model.features[40].weight_mask
sparsity_mask1 = (mask1 == 0).sum() / mask1.nelement()

print("Sparsity level: ", sparsity_mask1)

```

Sparsity level: tensor(0.8000, device='cuda:0')

```

[8]: model.cuda()
model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 1003/10000 (10%)

```

[9]: lr = 1e-1
weight_decay = 1e-4
epochs = 100
best_prec = 0

#model = nn.DataParallel(model).cuda()
model.cuda()
criterion = nn.CrossEntropyLoss().cuda()
optimizer = torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9,
    ↪weight_decay=weight_decay)
# cudnn.benchmark = True

if not os.path.exists('result'):
    os.makedirs('result')
fdir = 'result/'+'pruning'+str(model_name)
if not os.path.exists(fdir):
    os.makedirs(fdir)

for epoch in range(0, epochs):
    adjust_learning_rate(optimizer, epoch)

    train(trainloader, model, criterion, optimizer, epoch)

    # evaluate on test set
    print("Validation starts")
    prec = validate(testloader, model, criterion)

    # remember best precision and save checkpoint
    is_best = prec > best_prec
    best_prec = max(prec, best_prec)
    print('best acc: {:.1f}'.format(best_prec))
    save_checkpoint({
        'epoch': epoch + 1,
        'state_dict': model.state_dict(),
        'best_prec': best_prec,
        'optimizer': optimizer.state_dict(),
    }, is_best, fdir)

```

Epoch: [0] [0/391]	Time 0.283 (0.283)	Data 0.098 (0.098)	Loss
3.2798 (3.2798)	Prec 38.281% (38.281%)		
Epoch: [0] [100/391]	Time 0.056 (0.058)	Data 0.002 (0.002)	Loss
0.2768 (0.4462)	Prec 90.625% (86.177%)		
Epoch: [0] [200/391]	Time 0.057 (0.057)	Data 0.002 (0.002)	Loss
0.1193 (0.3433)	Prec 94.531% (89.066%)		
Epoch: [0] [300/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.1640 (0.2979)	Prec 94.531% (90.355%)		

Validation starts

Test: [0/79] Time 0.111 (0.111) Loss 0.8878 (0.8878) Prec 72.656%
(72.656%)

* Prec 72.370%

best acc: 72.370000

Epoch: [1][0/391]	Time 0.186 (0.186)	Data 0.140 (0.140)	Loss
0.0757 (0.0757)	Prec 99.219% (99.219%)		
Epoch: [1][100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.1474 (0.1504)	Prec 93.750% (95.034%)		
Epoch: [1][200/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.1729 (0.1452)	Prec 93.750% (95.145%)		
Epoch: [1][300/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.1193 (0.1447)	Prec 96.094% (95.102%)		

Validation starts

Test: [0/79] Time 0.118 (0.118) Loss 0.4078 (0.4078) Prec 89.062%
(89.062%)

* Prec 87.860%

best acc: 87.860000

Epoch: [2][0/391]	Time 0.156 (0.156)	Data 0.110 (0.110)	Loss
0.0856 (0.0856)	Prec 96.875% (96.875%)		
Epoch: [2][100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.1529 (0.1209)	Prec 93.750% (95.815%)		
Epoch: [2][200/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.1870 (0.1311)	Prec 95.312% (95.530%)		
Epoch: [2][300/391]	Time 0.055 (0.056)	Data 0.002 (0.002)	Loss
0.2585 (0.1344)	Prec 91.406% (95.427%)		

Validation starts

Test: [0/79] Time 0.108 (0.108) Loss 0.2295 (0.2295) Prec 92.188%
(92.188%)

* Prec 88.920%

best acc: 88.920000

Epoch: [3][0/391]	Time 0.199 (0.199)	Data 0.145 (0.145)	Loss
0.0673 (0.0673)	Prec 98.438% (98.438%)		
Epoch: [3][100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.1559 (0.1137)	Prec 95.312% (95.962%)		
Epoch: [3][200/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.0260 (0.1174)	Prec 100.000% (95.806%)		
Epoch: [3][300/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.1411 (0.1176)	Prec 94.531% (95.865%)		

Validation starts

Test: [0/79] Time 0.127 (0.127) Loss 0.2712 (0.2712) Prec 92.969%
(92.969%)

* Prec 89.040%

best acc: 89.040000

Epoch: [4][0/391]	Time 0.185 (0.185)	Data 0.140 (0.140)	Loss
0.0900 (0.0900)	Prec 96.875% (96.875%)		
Epoch: [4][100/391]	Time 0.057 (0.057)	Data 0.002 (0.003)	Loss
0.2355 (0.1069)	Prec 90.625% (96.163%)		

Epoch: [4][200/391] Time 0.057 (0.057) Data 0.002 (0.002) Loss
0.1395 (0.1075) Prec 96.094% (96.230%)
Epoch: [4][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1062 (0.1106) Prec 96.875% (96.127%)
Validation starts
Test: [0/79] Time 0.134 (0.134) Loss 0.2848 (0.2848) Prec 89.844%
(89.844%)
* Prec 89.790%
best acc: 89.790000

Epoch: [5][0/391] Time 0.207 (0.207) Data 0.157 (0.157) Loss
0.1079 (0.1079) Prec 96.875% (96.875%)
Epoch: [5][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.2246 (0.1054) Prec 92.188% (96.581%)
Epoch: [5][200/391] Time 0.062 (0.056) Data 0.001 (0.002) Loss
0.1990 (0.1101) Prec 93.750% (96.269%)
Epoch: [5][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1740 (0.1119) Prec 92.188% (96.190%)
Validation starts
Test: [0/79] Time 0.112 (0.112) Loss 0.3582 (0.3582) Prec 91.406%
(91.406%)
* Prec 88.220%
best acc: 89.790000

Epoch: [6][0/391] Time 0.186 (0.186) Data 0.143 (0.143) Loss
0.1251 (0.1251) Prec 96.094% (96.094%)
Epoch: [6][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.0680 (0.1018) Prec 98.438% (96.535%)
Epoch: [6][200/391] Time 0.056 (0.056) Data 0.002 (0.003) Loss
0.1246 (0.1052) Prec 96.875% (96.269%)
Epoch: [6][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1436 (0.1049) Prec 96.094% (96.283%)
Validation starts
Test: [0/79] Time 0.122 (0.122) Loss 0.2509 (0.2509) Prec 92.969%
(92.969%)
* Prec 89.010%
best acc: 89.790000

Epoch: [7][0/391] Time 0.194 (0.194) Data 0.139 (0.139) Loss
0.0998 (0.0998) Prec 96.875% (96.875%)
Epoch: [7][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.0736 (0.1037) Prec 96.094% (96.473%)
Epoch: [7][200/391] Time 0.055 (0.057) Data 0.002 (0.002) Loss
0.1390 (0.1086) Prec 95.312% (96.210%)
Epoch: [7][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1232 (0.1073) Prec 96.094% (96.231%)
Validation starts
Test: [0/79] Time 0.114 (0.114) Loss 0.3054 (0.3054) Prec 91.406%
(91.406%)
* Prec 88.840%
best acc: 89.790000

Epoch: [8][0/391] Time 0.182 (0.182) Data 0.141 (0.141) Loss
0.1458 (0.1458) Prec 95.312% (95.312%)

Epoch: [8][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.0962 (0.1078) Prec 96.094% (96.264%)

Epoch: [8][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1253 (0.1063) Prec 97.656% (96.265%)

Epoch: [8][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0588 (0.1069) Prec 98.438% (96.234%)

Validation starts
Test: [0/79] Time 0.117 (0.117) Loss 0.2191 (0.2191) Prec 92.188%
(92.188%)
* Prec 89.280%
best acc: 89.790000

Epoch: [9][0/391] Time 0.227 (0.227) Data 0.173 (0.173) Loss
0.0397 (0.0397) Prec 98.438% (98.438%)

Epoch: [9][100/391] Time 0.056 (0.058) Data 0.002 (0.003) Loss
0.0604 (0.0984) Prec 97.656% (96.473%)

Epoch: [9][200/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1065 (0.0970) Prec 97.656% (96.560%)

Epoch: [9][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1305 (0.1005) Prec 96.094% (96.426%)

Validation starts
Test: [0/79] Time 0.128 (0.128) Loss 0.2061 (0.2061) Prec 94.531%
(94.531%)
* Prec 88.880%
best acc: 89.790000

Epoch: [10][0/391] Time 0.174 (0.174) Data 0.135 (0.135) Loss
0.0952 (0.0952) Prec 95.312% (95.312%)

Epoch: [10][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.1297 (0.1028) Prec 96.094% (96.403%)

Epoch: [10][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.0741 (0.1049) Prec 97.656% (96.319%)

Epoch: [10][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0780 (0.1069) Prec 97.656% (96.244%)

Validation starts
Test: [0/79] Time 0.114 (0.114) Loss 0.2571 (0.2571) Prec 90.625%
(90.625%)
* Prec 89.550%
best acc: 89.790000

Epoch: [11][0/391] Time 0.185 (0.185) Data 0.130 (0.130) Loss
0.0942 (0.0942) Prec 95.312% (95.312%)

Epoch: [11][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1115 (0.0903) Prec 95.312% (96.535%)

Epoch: [11][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1211 (0.0932) Prec 95.312% (96.583%)

Epoch: [11][300/391] Time 0.056 (0.056) Data 0.011 (0.002) Loss
0.0915 (0.0985) Prec 96.875% (96.460%)

Validation starts

Test: [0/79] Time 0.138 (0.138) Loss 0.3979 (0.3979) Prec 89.062%
(89.062%)

* Prec 87.970%

best acc: 89.790000

Epoch: [12][0/391] Time 0.183 (0.183) Data 0.134 (0.134) Loss
0.0813 (0.0813) Prec 97.656% (97.656%)

Epoch: [12][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.0811 (0.0934) Prec 98.438% (96.829%)

Epoch: [12][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0608 (0.0990) Prec 99.219% (96.685%)

Epoch: [12][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1202 (0.1042) Prec 96.094% (96.434%)

Validation starts

Test: [0/79] Time 0.109 (0.109) Loss 0.3352 (0.3352) Prec 89.844%
(89.844%)

* Prec 87.650%

best acc: 89.790000

Epoch: [13][0/391] Time 0.185 (0.185) Data 0.138 (0.138) Loss
0.0996 (0.0996) Prec 97.656% (97.656%)

Epoch: [13][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.2470 (0.1034) Prec 92.969% (96.349%)

Epoch: [13][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1630 (0.1029) Prec 93.750% (96.366%)

Epoch: [13][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1326 (0.1048) Prec 97.656% (96.317%)

Validation starts

Test: [0/79] Time 0.117 (0.117) Loss 0.3850 (0.3850) Prec 90.625%
(90.625%)

* Prec 87.540%

best acc: 89.790000

Epoch: [14][0/391] Time 0.213 (0.213) Data 0.165 (0.165) Loss
0.0941 (0.0941) Prec 95.312% (95.312%)

Epoch: [14][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.1037 (0.1043) Prec 95.312% (96.364%)

Epoch: [14][200/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.0643 (0.1075) Prec 96.875% (96.296%)

Epoch: [14][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.1096 (0.1141) Prec 96.875% (96.008%)

Validation starts

Test: [0/79] Time 0.129 (0.129) Loss 0.2960 (0.2960) Prec 92.188%
(92.188%)

* Prec 88.910%

best acc: 89.790000

Epoch: [15][0/391] Time 0.168 (0.168) Data 0.124 (0.124) Loss
0.0876 (0.0876) Prec 99.219% (99.219%)

Epoch: [15][100/391] Time 0.058 (0.056) Data 0.001 (0.003) Loss
0.0441 (0.0917) Prec 97.656% (96.867%)

Epoch: [15][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss

0.1188 (0.0997) Prec 96.875% (96.459%)
Epoch: [15][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1509 (0.1066) Prec 92.969% (96.242%)
Validation starts
Test: [0/79] Time 0.106 (0.106) Loss 0.2878 (0.2878) Prec 89.844%
(89.844%)
* Prec 88.880%
best acc: 89.790000
Epoch: [16][0/391] Time 0.169 (0.169) Data 0.130 (0.130) Loss
0.1648 (0.1648) Prec 96.875% (96.875%)
Epoch: [16][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1468 (0.0922) Prec 95.312% (96.674%)
Epoch: [16][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1395 (0.0969) Prec 95.312% (96.502%)
Epoch: [16][300/391] Time 0.049 (0.056) Data 0.002 (0.002) Loss
0.1283 (0.1032) Prec 97.656% (96.322%)
Validation starts
Test: [0/79] Time 0.109 (0.109) Loss 0.3877 (0.3877) Prec 88.281%
(88.281%)
* Prec 89.270%
best acc: 89.790000
Epoch: [17][0/391] Time 0.194 (0.194) Data 0.143 (0.143) Loss
0.1060 (0.1060) Prec 96.094% (96.094%)
Epoch: [17][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.1045 (0.1039) Prec 96.094% (96.256%)
Epoch: [17][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0667 (0.1011) Prec 97.656% (96.428%)
Epoch: [17][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1099 (0.1049) Prec 95.312% (96.327%)
Validation starts
Test: [0/79] Time 0.126 (0.126) Loss 0.2702 (0.2702) Prec 88.281%
(88.281%)
* Prec 88.150%
best acc: 89.790000
Epoch: [18][0/391] Time 0.185 (0.185) Data 0.144 (0.144) Loss
0.1275 (0.1275) Prec 94.531% (94.531%)
Epoch: [18][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1474 (0.0994) Prec 93.750% (96.450%)
Epoch: [18][200/391] Time 0.055 (0.057) Data 0.002 (0.002) Loss
0.1253 (0.1021) Prec 95.312% (96.381%)
Epoch: [18][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.0655 (0.1050) Prec 97.656% (96.320%)
Validation starts
Test: [0/79] Time 0.129 (0.129) Loss 0.2692 (0.2692) Prec 90.625%
(90.625%)
* Prec 89.810%
best acc: 89.810000
Epoch: [19][0/391] Time 0.196 (0.196) Data 0.154 (0.154) Loss

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0.0965 (0.0965)    Prec 96.094% (96.094%)
Epoch: [19][100/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1204 (0.1077)    Prec 96.875% (96.256%)
Epoch: [19][200/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1362 (0.1051)    Prec 96.875% (96.412%)
Epoch: [19][300/391]    Time 0.055 (0.056)    Data 0.001 (0.002)    Loss
0.0960 (0.1052)    Prec 96.875% (96.356%)
Validation starts
Test: [0/79]    Time 0.119 (0.119)    Loss 0.2975 (0.2975)    Prec 90.625%
(90.625%)
* Prec 89.160%
best acc: 89.810000
Epoch: [20][0/391]    Time 0.188 (0.188)    Data 0.142 (0.142)    Loss
0.0966 (0.0966)    Prec 96.094% (96.094%)
Epoch: [20][100/391]    Time 0.057 (0.057)    Data 0.002 (0.003)    Loss
0.0652 (0.1078)    Prec 97.656% (96.117%)
Epoch: [20][200/391]    Time 0.056 (0.057)    Data 0.001 (0.002)    Loss
0.0991 (0.1098)    Prec 96.875% (96.059%)
Epoch: [20][300/391]    Time 0.056 (0.056)    Data 0.002 (0.002)    Loss
0.1138 (0.1110)    Prec 94.531% (96.065%)
Validation starts
Test: [0/79]    Time 0.127 (0.127)    Loss 0.2968 (0.2968)    Prec 88.281%
(88.281%)
* Prec 88.310%
best acc: 89.810000
Epoch: [21][0/391]    Time 0.196 (0.196)    Data 0.146 (0.146)    Loss
0.0634 (0.0634)    Prec 97.656% (97.656%)
Epoch: [21][100/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.0963 (0.0997)    Prec 95.312% (96.511%)
Epoch: [21][200/391]    Time 0.055 (0.057)    Data 0.002 (0.002)    Loss
0.0590 (0.1034)    Prec 96.875% (96.366%)
Epoch: [21][300/391]    Time 0.055 (0.056)    Data 0.002 (0.002)    Loss
0.1941 (0.1070)    Prec 92.969% (96.200%)
Validation starts
Test: [0/79]    Time 0.115 (0.115)    Loss 0.3051 (0.3051)    Prec 90.625%
(90.625%)
* Prec 89.130%
best acc: 89.810000
Epoch: [22][0/391]    Time 0.168 (0.168)    Data 0.128 (0.128)    Loss
0.0694 (0.0694)    Prec 97.656% (97.656%)
Epoch: [22][100/391]    Time 0.055 (0.057)    Data 0.001 (0.003)    Loss
0.1377 (0.1037)    Prec 95.312% (96.326%)
Epoch: [22][200/391]    Time 0.055 (0.056)    Data 0.001 (0.002)    Loss
0.0580 (0.1050)    Prec 98.438% (96.304%)
Epoch: [22][300/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.0531 (0.1083)    Prec 96.875% (96.216%)
Validation starts
Test: [0/79]    Time 0.112 (0.112)    Loss 0.3435 (0.3435)    Prec 92.188%

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(92.188%)
* Prec 88.920%
best acc: 89.810000
Epoch: [23][0/391]      Time 0.193 (0.193)      Data 0.152 (0.152)      Loss
0.1327 (0.1327)      Prec 96.875% (96.875%)
Epoch: [23][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.0719 (0.0901)      Prec 96.875% (96.883%)
Epoch: [23][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.0780 (0.0966)      Prec 99.219% (96.646%)
Epoch: [23][300/391]    Time 0.051 (0.056)      Data 0.001 (0.002)      Loss
0.1527 (0.1021)      Prec 96.094% (96.478%)
Validation starts
Test: [0/79]      Time 0.117 (0.117)      Loss 0.3508 (0.3508)      Prec 89.062%
(89.062%)
* Prec 89.130%
best acc: 89.810000
Epoch: [24][0/391]      Time 0.195 (0.195)      Data 0.141 (0.141)      Loss
0.2017 (0.2017)      Prec 92.188% (92.188%)
Epoch: [24][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.1442 (0.0998)      Prec 95.312% (96.488%)
Epoch: [24][200/391]    Time 0.056 (0.057)      Data 0.002 (0.002)      Loss
0.0548 (0.1062)      Prec 97.656% (96.234%)
Epoch: [24][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0670 (0.1102)      Prec 97.656% (96.096%)
Validation starts
Test: [0/79]      Time 0.114 (0.114)      Loss 0.2370 (0.2370)      Prec 92.188%
(92.188%)
* Prec 88.940%
best acc: 89.810000
Epoch: [25][0/391]      Time 0.186 (0.186)      Data 0.135 (0.135)      Loss
0.0557 (0.0557)      Prec 97.656% (97.656%)
Epoch: [25][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0567 (0.0967)      Prec 98.438% (96.651%)
Epoch: [25][200/391]    Time 0.056 (0.057)      Data 0.002 (0.002)      Loss
0.0857 (0.1008)      Prec 97.656% (96.533%)
Epoch: [25][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.1801 (0.1054)      Prec 94.531% (96.434%)
Validation starts
Test: [0/79]      Time 0.125 (0.125)      Loss 0.2816 (0.2816)      Prec 89.844%
(89.844%)
* Prec 89.570%
best acc: 89.810000
Epoch: [26][0/391]      Time 0.188 (0.188)      Data 0.138 (0.138)      Loss
0.0596 (0.0596)      Prec 98.438% (98.438%)
Epoch: [26][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0666 (0.0915)      Prec 98.438% (96.929%)
Epoch: [26][200/391]    Time 0.055 (0.057)      Data 0.002 (0.002)      Loss
0.1689 (0.0997)      Prec 94.531% (96.568%)

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Epoch: [26][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1341 (0.1048) Prec 95.312% (96.322%)
Validation starts
Test: [0/79] Time 0.125 (0.125) Loss 0.3853 (0.3853) Prec 90.625%
(90.625%)
* Prec 89.070%
best acc: 89.810000
Epoch: [27][0/391] Time 0.184 (0.184) Data 0.138 (0.138) Loss
0.0648 (0.0648) Prec 97.656% (97.656%)
Epoch: [27][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.2077 (0.1032) Prec 93.750% (96.395%)
Epoch: [27][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1142 (0.1000) Prec 97.656% (96.591%)
Epoch: [27][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0656 (0.0975) Prec 96.875% (96.654%)
Validation starts
Test: [0/79] Time 0.131 (0.131) Loss 0.4218 (0.4218) Prec 85.938%
(85.938%)
* Prec 88.980%
best acc: 89.810000
Epoch: [28][0/391] Time 0.219 (0.219) Data 0.166 (0.166) Loss
0.0935 (0.0935) Prec 95.312% (95.312%)
Epoch: [28][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1103 (0.1075) Prec 95.312% (96.295%)
Epoch: [28][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0738 (0.1139) Prec 96.875% (95.938%)
Epoch: [28][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0584 (0.1128) Prec 98.438% (96.024%)
Validation starts
Test: [0/79] Time 0.113 (0.113) Loss 0.3702 (0.3702) Prec 90.625%
(90.625%)
* Prec 88.490%
best acc: 89.810000
Epoch: [29][0/391] Time 0.191 (0.191) Data 0.143 (0.143) Loss
0.1005 (0.1005) Prec 96.875% (96.875%)
Epoch: [29][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.0291 (0.1053) Prec 98.438% (96.218%)
Epoch: [29][200/391] Time 0.062 (0.056) Data 0.002 (0.002) Loss
0.0939 (0.1058) Prec 97.656% (96.280%)
Epoch: [29][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0733 (0.1055) Prec 96.875% (96.351%)
Validation starts
Test: [0/79] Time 0.122 (0.122) Loss 0.4933 (0.4933) Prec 86.719%
(86.719%)
* Prec 87.610%
best acc: 89.810000
Epoch: [30][0/391] Time 0.181 (0.181) Data 0.133 (0.133) Loss
0.1844 (0.1844) Prec 95.312% (95.312%)

Epoch: [30][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.0748 (0.1059) Prec 96.875% (96.341%)
 Epoch: [30][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.0540 (0.1094) Prec 98.438% (96.222%)
 Epoch: [30][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.0919 (0.1086) Prec 96.094% (96.320%)
 Validation starts
 Test: [0/79] Time 0.114 (0.114) Loss 0.1924 (0.1924) Prec 93.750%
 (93.750%)
 * Prec 89.560%
 best acc: 89.810000
 Epoch: [31][0/391] Time 0.192 (0.192) Data 0.143 (0.143) Loss
 0.0405 (0.0405) Prec 99.219% (99.219%)
 Epoch: [31][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
 0.0690 (0.0938) Prec 97.656% (96.860%)
 Epoch: [31][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1825 (0.1004) Prec 93.750% (96.471%)
 Epoch: [31][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1234 (0.1048) Prec 97.656% (96.382%)
 Validation starts
 Test: [0/79] Time 0.130 (0.130) Loss 0.3922 (0.3922) Prec 88.281%
 (88.281%)
 * Prec 88.330%
 best acc: 89.810000
 Epoch: [32][0/391] Time 0.178 (0.178) Data 0.137 (0.137) Loss
 0.0680 (0.0680) Prec 96.875% (96.875%)
 Epoch: [32][100/391] Time 0.057 (0.057) Data 0.002 (0.003) Loss
 0.0660 (0.0934) Prec 98.438% (96.713%)
 Epoch: [32][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.1183 (0.0990) Prec 95.312% (96.541%)
 Epoch: [32][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.0851 (0.1019) Prec 98.438% (96.418%)
 Validation starts
 Test: [0/79] Time 0.113 (0.113) Loss 0.3158 (0.3158) Prec 89.844%
 (89.844%)
 * Prec 88.580%
 best acc: 89.810000
 Epoch: [33][0/391] Time 0.170 (0.170) Data 0.132 (0.132) Loss
 0.1121 (0.1121) Prec 95.312% (95.312%)
 Epoch: [33][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.0669 (0.1092) Prec 97.656% (96.210%)
 Epoch: [33][200/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
 0.1609 (0.1080) Prec 92.969% (96.226%)
 Epoch: [33][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
 0.1623 (0.1123) Prec 93.750% (96.096%)
 Validation starts
 Test: [0/79] Time 0.121 (0.121) Loss 0.2231 (0.2231) Prec 91.406%
 (91.406%)

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* Prec 88.610%
best acc: 89.810000
Epoch: [34][0/391]      Time 0.191 (0.191)      Data 0.142 (0.142)      Loss
0.1075 (0.1075)      Prec 95.312% (95.312%)
Epoch: [34][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1950 (0.1033)      Prec 91.406% (96.303%)
Epoch: [34][200/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1164 (0.1005)      Prec 93.750% (96.381%)
Epoch: [34][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.1224 (0.1040)      Prec 96.094% (96.286%)
Validation starts
Test: [0/79]      Time 0.105 (0.105)      Loss 0.3161 (0.3161)      Prec 91.406%
(91.406%)
* Prec 87.670%
best acc: 89.810000
Epoch: [35][0/391]      Time 0.179 (0.179)      Data 0.128 (0.128)      Loss
0.1005 (0.1005)      Prec 96.094% (96.094%)
Epoch: [35][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1048 (0.1031)      Prec 94.531% (96.364%)
Epoch: [35][200/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.0595 (0.1075)      Prec 98.438% (96.187%)
Epoch: [35][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1597 (0.1127)      Prec 94.531% (96.024%)
Validation starts
Test: [0/79]      Time 0.105 (0.105)      Loss 0.3456 (0.3456)      Prec 88.281%
(88.281%)
* Prec 88.120%
best acc: 89.810000
Epoch: [36][0/391]      Time 0.187 (0.187)      Data 0.146 (0.146)      Loss
0.0763 (0.0763)      Prec 96.875% (96.875%)
Epoch: [36][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1145 (0.0997)      Prec 95.312% (96.473%)
Epoch: [36][200/391]    Time 0.056 (0.057)      Data 0.001 (0.002)      Loss
0.0555 (0.1052)      Prec 99.219% (96.234%)
Epoch: [36][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0495 (0.1042)      Prec 99.219% (96.353%)
Validation starts
Test: [0/79]      Time 0.128 (0.128)      Loss 0.1450 (0.1450)      Prec 96.094%
(96.094%)
* Prec 88.990%
best acc: 89.810000
Epoch: [37][0/391]      Time 0.204 (0.204)      Data 0.156 (0.156)      Loss
0.0850 (0.0850)      Prec 98.438% (98.438%)
Epoch: [37][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0713 (0.0977)      Prec 97.656% (96.496%)
Epoch: [37][200/391]    Time 0.057 (0.057)      Data 0.002 (0.003)      Loss
0.1210 (0.1030)      Prec 95.312% (96.308%)
Epoch: [37][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss

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0.0920 (0.1080) Prec 96.875% (96.161%)
Validation starts
Test: [0/79] Time 0.130 (0.130) Loss 0.3619 (0.3619) Prec 90.625% (90.625%)
* Prec 89.550%
best acc: 89.810000
Epoch: [38][0/391] Time 0.179 (0.179) Data 0.131 (0.131) Loss 0.1805 (0.1805) Prec 94.531% (94.531%)
Epoch: [38][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss 0.0886 (0.1040) Prec 96.094% (96.279%)
Epoch: [38][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss 0.0940 (0.1034) Prec 96.094% (96.335%)
Epoch: [38][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss 0.2247 (0.1054) Prec 92.969% (96.231%)
Validation starts
Test: [0/79] Time 0.109 (0.109) Loss 0.2279 (0.2279) Prec 92.969% (92.969%)
* Prec 89.660%
best acc: 89.810000
Epoch: [39][0/391] Time 0.160 (0.160) Data 0.120 (0.120) Loss 0.0863 (0.0863) Prec 96.875% (96.875%)
Epoch: [39][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss 0.0985 (0.1135) Prec 96.094% (95.939%)
Epoch: [39][200/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss 0.1682 (0.1086) Prec 93.750% (96.203%)
Epoch: [39][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss 0.1933 (0.1105) Prec 94.531% (96.177%)
Validation starts
Test: [0/79] Time 0.109 (0.109) Loss 0.2353 (0.2353) Prec 93.750% (93.750%)
* Prec 89.100%
best acc: 89.810000
Epoch: [40][0/391] Time 0.172 (0.172) Data 0.132 (0.132) Loss 0.0737 (0.0737) Prec 97.656% (97.656%)
Epoch: [40][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss 0.1561 (0.1004) Prec 92.969% (96.581%)
Epoch: [40][200/391] Time 0.059 (0.056) Data 0.002 (0.002) Loss 0.0958 (0.1022) Prec 96.875% (96.440%)
Epoch: [40][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss 0.1744 (0.1053) Prec 93.750% (96.278%)
Validation starts
Test: [0/79] Time 0.115 (0.115) Loss 0.2952 (0.2952) Prec 93.750% (93.750%)
* Prec 89.620%
best acc: 89.810000
Epoch: [41][0/391] Time 0.200 (0.200) Data 0.158 (0.158) Loss 0.0629 (0.0629) Prec 96.875% (96.875%)
Epoch: [41][100/391] Time 0.058 (0.057) Data 0.002 (0.003) Loss

0.0169 (0.1010) Prec 100.000% (96.566%)
 Epoch: [41][200/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.0474 (0.1061) Prec 100.000% (96.463%)
 Epoch: [41][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.0519 (0.1081) Prec 97.656% (96.281%)
 Validation starts
 Test: [0/79] Time 0.110 (0.110) Loss 0.2341 (0.2341) Prec 89.844%
 (89.844%)
 * Prec 88.860%
 best acc: 89.810000
 Epoch: [42][0/391] Time 0.174 (0.174) Data 0.134 (0.134) Loss
 0.1035 (0.1035) Prec 95.312% (95.312%)
 Epoch: [42][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.0527 (0.0963) Prec 97.656% (96.713%)
 Epoch: [42][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.0343 (0.0993) Prec 98.438% (96.502%)
 Epoch: [42][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.0651 (0.1031) Prec 98.438% (96.413%)
 Validation starts
 Test: [0/79] Time 0.121 (0.121) Loss 0.3481 (0.3481) Prec 88.281%
 (88.281%)
 * Prec 88.150%
 best acc: 89.810000
 Epoch: [43][0/391] Time 0.187 (0.187) Data 0.139 (0.139) Loss
 0.0750 (0.0750) Prec 97.656% (97.656%)
 Epoch: [43][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
 0.0872 (0.1059) Prec 96.094% (96.326%)
 Epoch: [43][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.2219 (0.1105) Prec 92.969% (96.187%)
 Epoch: [43][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.1504 (0.1105) Prec 94.531% (96.198%)
 Validation starts
 Test: [0/79] Time 0.132 (0.132) Loss 0.2345 (0.2345) Prec 93.750%
 (93.750%)
 * Prec 88.430%
 best acc: 89.810000
 Epoch: [44][0/391] Time 0.185 (0.185) Data 0.136 (0.136) Loss
 0.1352 (0.1352) Prec 94.531% (94.531%)
 Epoch: [44][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.1245 (0.1024) Prec 95.312% (96.457%)
 Epoch: [44][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.0962 (0.1051) Prec 96.094% (96.288%)
 Epoch: [44][300/391] Time 0.058 (0.056) Data 0.001 (0.002) Loss
 0.0425 (0.1079) Prec 98.438% (96.198%)
 Validation starts
 Test: [0/79] Time 0.114 (0.114) Loss 0.2602 (0.2602) Prec 90.625%
 (90.625%)
 * Prec 88.950%


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best acc: 89.810000
Epoch: [45][0/391]      Time 0.197 (0.197)      Data 0.148 (0.148)      Loss
0.0796 (0.0796)      Prec 96.875% (96.875%)
Epoch: [45][100/391]    Time 0.057 (0.057)      Data 0.001 (0.003)      Loss
0.1440 (0.1065)      Prec 96.094% (96.117%)
Epoch: [45][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.0433 (0.1120)      Prec 99.219% (95.973%)
Epoch: [45][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1507 (0.1122)      Prec 94.531% (96.044%)
Validation starts
Test: [0/79]      Time 0.116 (0.116)      Loss 0.3284 (0.3284)      Prec 89.844%
(89.844%)
* Prec 88.130%
best acc: 89.810000
Epoch: [46][0/391]      Time 0.176 (0.176)      Data 0.134 (0.134)      Loss
0.0453 (0.0453)      Prec 99.219% (99.219%)
Epoch: [46][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0900 (0.1021)      Prec 97.656% (96.388%)
Epoch: [46][200/391]    Time 0.056 (0.057)      Data 0.002 (0.002)      Loss
0.0899 (0.1041)      Prec 96.875% (96.350%)
Epoch: [46][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2413 (0.1060)      Prec 93.750% (96.203%)
Validation starts
Test: [0/79]      Time 0.138 (0.138)      Loss 0.2853 (0.2853)      Prec 92.188%
(92.188%)
* Prec 90.330%
best acc: 90.330000
Epoch: [47][0/391]      Time 0.194 (0.194)      Data 0.144 (0.144)      Loss
0.0733 (0.0733)      Prec 96.094% (96.094%)
Epoch: [47][100/391]    Time 0.057 (0.057)      Data 0.002 (0.003)      Loss
0.1162 (0.1002)      Prec 93.750% (96.372%)
Epoch: [47][200/391]    Time 0.056 (0.057)      Data 0.001 (0.002)      Loss
0.0884 (0.1041)      Prec 96.875% (96.358%)
Epoch: [47][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1068 (0.1045)      Prec 95.312% (96.361%)
Validation starts
Test: [0/79]      Time 0.107 (0.107)      Loss 0.2428 (0.2428)      Prec 91.406%
(91.406%)
* Prec 87.680%
best acc: 90.330000
Epoch: [48][0/391]      Time 0.169 (0.169)      Data 0.127 (0.127)      Loss
0.0781 (0.0781)      Prec 97.656% (97.656%)
Epoch: [48][100/391]    Time 0.055 (0.057)      Data 0.001 (0.003)      Loss
0.0926 (0.0972)      Prec 96.875% (96.519%)
Epoch: [48][200/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0496 (0.0982)      Prec 98.438% (96.521%)
Epoch: [48][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1071 (0.1022)      Prec 96.875% (96.442%)

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Validation starts

Test: [0/79] Time 0.115 (0.115) Loss 0.2177 (0.2177) Prec 92.188%
(92.188%)

* Prec 89.840%

best acc: 90.330000

Epoch: [49] [0/391]	Time 0.171 (0.171)	Data 0.122 (0.122)	Loss
0.1174 (0.1174)	Prec 96.094% (96.094%)		
Epoch: [49] [100/391]	Time 0.056 (0.057)	Data 0.001 (0.003)	Loss
0.0704 (0.1057)	Prec 96.875% (96.364%)		
Epoch: [49] [200/391]	Time 0.057 (0.056)	Data 0.001 (0.002)	Loss
0.1006 (0.1070)	Prec 96.875% (96.319%)		
Epoch: [49] [300/391]	Time 0.057 (0.056)	Data 0.001 (0.002)	Loss
0.1177 (0.1122)	Prec 96.875% (96.078%)		

Validation starts

Test: [0/79] Time 0.122 (0.122) Loss 0.2574 (0.2574) Prec 93.750%
(93.750%)

* Prec 88.610%

best acc: 90.330000

Epoch: [50] [0/391]	Time 0.175 (0.175)	Data 0.127 (0.127)	Loss
0.1135 (0.1135)	Prec 95.312% (95.312%)		
Epoch: [50] [100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.0962 (0.1044)	Prec 96.875% (96.357%)		
Epoch: [50] [200/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.1340 (0.1034)	Prec 96.094% (96.393%)		
Epoch: [50] [300/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.1244 (0.1068)	Prec 96.094% (96.281%)		

Validation starts

Test: [0/79] Time 0.105 (0.105) Loss 0.3316 (0.3316) Prec 92.188%
(92.188%)

* Prec 88.810%

best acc: 90.330000

Epoch: [51] [0/391]	Time 0.192 (0.192)	Data 0.148 (0.148)	Loss
0.0852 (0.0852)	Prec 96.875% (96.875%)		
Epoch: [51] [100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.0818 (0.0984)	Prec 97.656% (96.589%)		
Epoch: [51] [200/391]	Time 0.056 (0.057)	Data 0.002 (0.002)	Loss
0.1604 (0.1031)	Prec 92.969% (96.374%)		
Epoch: [51] [300/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.1408 (0.1095)	Prec 95.312% (96.143%)		

Validation starts

Test: [0/79] Time 0.139 (0.139) Loss 0.2444 (0.2444) Prec 93.750%
(93.750%)

* Prec 89.210%

best acc: 90.330000

Epoch: [52] [0/391]	Time 0.189 (0.189)	Data 0.143 (0.143)	Loss
0.0669 (0.0669)	Prec 98.438% (98.438%)		
Epoch: [52] [100/391]	Time 0.062 (0.057)	Data 0.002 (0.003)	Loss
0.0802 (0.1145)	Prec 96.094% (96.148%)		

Epoch: [52][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1444 (0.1138) Prec 96.094% (96.012%)

Epoch: [52][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1289 (0.1133) Prec 96.094% (95.990%)

Validation starts
Test: [0/79] Time 0.123 (0.123) Loss 0.2687 (0.2687) Prec 91.406%
(91.406%)
* Prec 88.990%
best acc: 90.330000

Epoch: [53][0/391] Time 0.186 (0.186) Data 0.137 (0.137) Loss
0.1043 (0.1043) Prec 97.656% (97.656%)

Epoch: [53][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1116 (0.1096) Prec 96.094% (96.233%)

Epoch: [53][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0965 (0.1117) Prec 96.094% (96.137%)

Epoch: [53][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0750 (0.1144) Prec 97.656% (95.993%)

Validation starts
Test: [0/79] Time 0.114 (0.114) Loss 0.3207 (0.3207) Prec 90.625%
(90.625%)
* Prec 88.280%
best acc: 90.330000

Epoch: [54][0/391] Time 0.169 (0.169) Data 0.129 (0.129) Loss
0.1642 (0.1642) Prec 94.531% (94.531%)

Epoch: [54][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.0687 (0.1078) Prec 97.656% (96.248%)

Epoch: [54][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1137 (0.1093) Prec 95.312% (96.257%)

Epoch: [54][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1650 (0.1092) Prec 93.750% (96.249%)

Validation starts
Test: [0/79] Time 0.106 (0.106) Loss 0.3660 (0.3660) Prec 90.625%
(90.625%)
* Prec 88.660%
best acc: 90.330000

Epoch: [55][0/391] Time 0.212 (0.212) Data 0.156 (0.156) Loss
0.0503 (0.0503) Prec 97.656% (97.656%)

Epoch: [55][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1367 (0.1072) Prec 96.094% (96.187%)

Epoch: [55][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.0887 (0.1101) Prec 97.656% (96.121%)

Epoch: [55][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1044 (0.1133) Prec 95.312% (96.003%)

Validation starts
Test: [0/79] Time 0.123 (0.123) Loss 0.4068 (0.4068) Prec 90.625%
(90.625%)
* Prec 89.320%
best acc: 90.330000

Epoch: [56][0/391] Time 0.185 (0.185) Data 0.137 (0.137) Loss
0.0966 (0.0966) Prec 96.094% (96.094%)

Epoch: [56][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1012 (0.1084) Prec 97.656% (96.187%)

Epoch: [56][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1593 (0.1176) Prec 95.312% (95.899%)

Epoch: [56][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0736 (0.1182) Prec 97.656% (95.826%)

Validation starts
Test: [0/79] Time 0.128 (0.128) Loss 0.2932 (0.2932) Prec 90.625%
(90.625%)
* Prec 88.980%
best acc: 90.330000

Epoch: [57][0/391] Time 0.200 (0.200) Data 0.148 (0.148) Loss
0.0697 (0.0697) Prec 98.438% (98.438%)

Epoch: [57][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.1331 (0.1108) Prec 96.094% (96.055%)

Epoch: [57][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1192 (0.1188) Prec 94.531% (95.748%)

Epoch: [57][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.0853 (0.1186) Prec 96.875% (95.751%)

Validation starts
Test: [0/79] Time 0.118 (0.118) Loss 0.3446 (0.3446) Prec 90.625%
(90.625%)
* Prec 89.490%
best acc: 90.330000

Epoch: [58][0/391] Time 0.190 (0.190) Data 0.142 (0.142) Loss
0.0479 (0.0479) Prec 98.438% (98.438%)

Epoch: [58][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.0917 (0.1110) Prec 96.094% (96.125%)

Epoch: [58][200/391] Time 0.054 (0.056) Data 0.001 (0.002) Loss
0.1230 (0.1103) Prec 96.094% (96.113%)

Epoch: [58][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1163 (0.1124) Prec 96.094% (96.065%)

Validation starts
Test: [0/79] Time 0.106 (0.106) Loss 0.3082 (0.3082) Prec 88.281%
(88.281%)
* Prec 87.920%
best acc: 90.330000

Epoch: [59][0/391] Time 0.187 (0.187) Data 0.138 (0.138) Loss
0.1969 (0.1969) Prec 94.531% (94.531%)

Epoch: [59][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.0916 (0.1127) Prec 97.656% (96.148%)

Epoch: [59][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1499 (0.1155) Prec 95.312% (96.063%)

Epoch: [59][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1001 (0.1187) Prec 96.094% (95.907%)

Validation starts

Test: [0/79] Time 0.142 (0.142) Loss 0.3133 (0.3133) Prec 94.531%
(94.531%)

* Prec 89.280%

best acc: 90.330000

Epoch: [60][0/391] Time 0.177 (0.177) Data 0.139 (0.139) Loss
0.1072 (0.1072) Prec 96.094% (96.094%)

Epoch: [60][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.1017 (0.1161) Prec 96.875% (95.978%)

Epoch: [60][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0895 (0.1136) Prec 96.875% (96.004%)

Epoch: [60][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.1722 (0.1181) Prec 92.969% (95.865%)

Validation starts

Test: [0/79] Time 0.123 (0.123) Loss 0.2484 (0.2484) Prec 89.844%
(89.844%)

* Prec 88.760%

best acc: 90.330000

Epoch: [61][0/391] Time 0.174 (0.174) Data 0.133 (0.133) Loss
0.0559 (0.0559) Prec 98.438% (98.438%)

Epoch: [61][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.2129 (0.0963) Prec 90.625% (96.349%)

Epoch: [61][200/391] Time 0.051 (0.056) Data 0.001 (0.002) Loss
0.1148 (0.1044) Prec 94.531% (96.121%)

Epoch: [61][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1064 (0.1084) Prec 96.875% (96.016%)

Validation starts

Test: [0/79] Time 0.114 (0.114) Loss 0.3084 (0.3084) Prec 89.844%
(89.844%)

* Prec 88.530%

best acc: 90.330000

Epoch: [62][0/391] Time 0.191 (0.191) Data 0.151 (0.151) Loss
0.1148 (0.1148) Prec 96.875% (96.875%)

Epoch: [62][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.0423 (0.1086) Prec 98.438% (96.109%)

Epoch: [62][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1785 (0.1121) Prec 93.750% (96.012%)

Epoch: [62][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1218 (0.1109) Prec 95.312% (96.107%)

Validation starts

Test: [0/79] Time 0.125 (0.125) Loss 0.3799 (0.3799) Prec 90.625%
(90.625%)

* Prec 89.570%

best acc: 90.330000

Epoch: [63][0/391] Time 0.221 (0.221) Data 0.166 (0.166) Loss
0.0914 (0.0914) Prec 97.656% (97.656%)

Epoch: [63][100/391] Time 0.056 (0.058) Data 0.002 (0.003) Loss
0.1387 (0.1124) Prec 93.750% (96.071%)

Epoch: [63][200/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss

0.0673 (0.1093) Prec 98.438% (96.144%)
 Epoch: [63][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1020 (0.1151) Prec 97.656% (95.967%)
 Validation starts
 Test: [0/79] Time 0.112 (0.112) Loss 0.3395 (0.3395) Prec 91.406%
 (91.406%)
 * Prec 89.210%
 best acc: 90.330000
 Epoch: [64][0/391] Time 0.193 (0.193) Data 0.145 (0.145) Loss
 0.1610 (0.1610) Prec 96.875% (96.875%)
 Epoch: [64][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.1687 (0.1044) Prec 93.750% (96.210%)
 Epoch: [64][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.0708 (0.1060) Prec 99.219% (96.199%)
 Epoch: [64][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.1061 (0.1064) Prec 95.312% (96.255%)
 Validation starts
 Test: [0/79] Time 0.124 (0.124) Loss 0.3093 (0.3093) Prec 92.969%
 (92.969%)
 * Prec 89.100%
 best acc: 90.330000
 Epoch: [65][0/391] Time 0.197 (0.197) Data 0.150 (0.150) Loss
 0.0767 (0.0767) Prec 96.875% (96.875%)
 Epoch: [65][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.1202 (0.1105) Prec 94.531% (95.962%)
 Epoch: [65][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.0802 (0.1080) Prec 96.875% (96.164%)
 Epoch: [65][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1433 (0.1063) Prec 92.969% (96.211%)
 Validation starts
 Test: [0/79] Time 0.116 (0.116) Loss 0.4771 (0.4771) Prec 90.625%
 (90.625%)
 * Prec 88.490%
 best acc: 90.330000
 Epoch: [66][0/391] Time 0.205 (0.205) Data 0.147 (0.147) Loss
 0.1237 (0.1237) Prec 95.312% (95.312%)
 Epoch: [66][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.1291 (0.1036) Prec 96.094% (96.465%)
 Epoch: [66][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.0392 (0.1073) Prec 99.219% (96.292%)
 Epoch: [66][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.0632 (0.1098) Prec 98.438% (96.182%)
 Validation starts
 Test: [0/79] Time 0.119 (0.119) Loss 0.3058 (0.3058) Prec 89.062%
 (89.062%)
 * Prec 88.860%
 best acc: 90.330000
 Epoch: [67][0/391] Time 0.174 (0.174) Data 0.135 (0.135) Loss

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0.1601 (0.1601)    Prec 94.531% (94.531%)
Epoch: [67][100/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1393 (0.0985)    Prec 95.312% (96.550%)
Epoch: [67][200/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.0799 (0.1083)    Prec 96.875% (96.199%)
Epoch: [67][300/391]    Time 0.055 (0.056)    Data 0.002 (0.002)    Loss
0.1020 (0.1082)    Prec 94.531% (96.211%)
Validation starts
Test: [0/79]    Time 0.110 (0.110)    Loss 0.3395 (0.3395)    Prec 89.844%
(89.844%)
* Prec 88.610%
best acc: 90.330000
Epoch: [68][0/391]    Time 0.204 (0.204)    Data 0.161 (0.161)    Loss
0.1711 (0.1711)    Prec 92.188% (92.188%)
Epoch: [68][100/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1519 (0.1037)    Prec 94.531% (96.419%)
Epoch: [68][200/391]    Time 0.055 (0.057)    Data 0.001 (0.002)    Loss
0.0656 (0.1063)    Prec 97.656% (96.284%)
Epoch: [68][300/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.2248 (0.1110)    Prec 93.750% (96.133%)
Validation starts
Test: [0/79]    Time 0.117 (0.117)    Loss 0.3744 (0.3744)    Prec 91.406%
(91.406%)
* Prec 88.630%
best acc: 90.330000
Epoch: [69][0/391]    Time 0.166 (0.166)    Data 0.118 (0.118)    Loss
0.1808 (0.1808)    Prec 93.750% (93.750%)
Epoch: [69][100/391]    Time 0.055 (0.057)    Data 0.002 (0.003)    Loss
0.1932 (0.1134)    Prec 93.750% (95.862%)
Epoch: [69][200/391]    Time 0.056 (0.056)    Data 0.002 (0.002)    Loss
0.0641 (0.1150)    Prec 97.656% (95.958%)
Epoch: [69][300/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.1544 (0.1149)    Prec 97.656% (95.961%)
Validation starts
Test: [0/79]    Time 0.105 (0.105)    Loss 0.3418 (0.3418)    Prec 92.188%
(92.188%)
* Prec 88.360%
best acc: 90.330000
Epoch: [70][0/391]    Time 0.180 (0.180)    Data 0.128 (0.128)    Loss
0.0616 (0.0616)    Prec 96.875% (96.875%)
Epoch: [70][100/391]    Time 0.055 (0.057)    Data 0.002 (0.003)    Loss
0.1830 (0.1025)    Prec 92.188% (96.194%)
Epoch: [70][200/391]    Time 0.055 (0.056)    Data 0.001 (0.002)    Loss
0.0970 (0.1098)    Prec 96.094% (96.008%)
Epoch: [70][300/391]    Time 0.055 (0.056)    Data 0.001 (0.002)    Loss
0.1787 (0.1104)    Prec 95.312% (96.037%)
Validation starts
Test: [0/79]    Time 0.115 (0.115)    Loss 0.2838 (0.2838)    Prec 92.188%

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(92.188%)

* Prec 88.850%

best acc: 90.330000

Epoch: [71][0/391]	Time 0.185 (0.185)	Data 0.138 (0.138)	Loss
0.1049 (0.1049)	Prec 95.312% (95.312%)		
Epoch: [71][100/391]	Time 0.055 (0.057)	Data 0.002 (0.003)	Loss
0.0532 (0.1055)	Prec 98.438% (96.210%)		
Epoch: [71][200/391]	Time 0.056 (0.057)	Data 0.002 (0.002)	Loss
0.0419 (0.1085)	Prec 99.219% (96.133%)		
Epoch: [71][300/391]	Time 0.055 (0.056)	Data 0.002 (0.002)	Loss
0.0993 (0.1089)	Prec 95.312% (96.169%)		

Validation starts

Test: [0/79] Time 0.136 (0.136) Loss 0.3941 (0.3941) Prec 89.062% (89.062%)

* Prec 88.430%

best acc: 90.330000

Epoch: [72][0/391]	Time 0.197 (0.197)	Data 0.149 (0.149)	Loss
0.0970 (0.0970)	Prec 97.656% (97.656%)		
Epoch: [72][100/391]	Time 0.056 (0.057)	Data 0.001 (0.003)	Loss
0.0892 (0.1139)	Prec 95.312% (96.132%)		
Epoch: [72][200/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.0984 (0.1122)	Prec 96.094% (96.109%)		
Epoch: [72][300/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.0743 (0.1137)	Prec 97.656% (96.039%)		

Validation starts

Test: [0/79] Time 0.112 (0.112) Loss 0.2954 (0.2954) Prec 89.844% (89.844%)

* Prec 89.020%

best acc: 90.330000

Epoch: [73][0/391]	Time 0.180 (0.180)	Data 0.130 (0.130)	Loss
0.0338 (0.0338)	Prec 99.219% (99.219%)		
Epoch: [73][100/391]	Time 0.056 (0.057)	Data 0.002 (0.003)	Loss
0.0753 (0.0979)	Prec 96.875% (96.682%)		
Epoch: [73][200/391]	Time 0.056 (0.057)	Data 0.002 (0.002)	Loss
0.1009 (0.1052)	Prec 96.094% (96.370%)		
Epoch: [73][300/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.2374 (0.1070)	Prec 93.750% (96.312%)		

Validation starts

Test: [0/79] Time 0.116 (0.116) Loss 0.4455 (0.4455) Prec 89.062% (89.062%)

* Prec 89.210%

best acc: 90.330000

Epoch: [74][0/391]	Time 0.192 (0.192)	Data 0.152 (0.152)	Loss
0.0773 (0.0773)	Prec 96.875% (96.875%)		
Epoch: [74][100/391]	Time 0.056 (0.057)	Data 0.001 (0.003)	Loss
0.0849 (0.1066)	Prec 96.094% (96.071%)		
Epoch: [74][200/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.2036 (0.1100)	Prec 93.750% (96.090%)		

Epoch: [74][300/391] Time 0.057 (0.056) Data 0.002 (0.002) Loss
0.1155 (0.1128) Prec 94.531% (96.050%)
Validation starts
Test: [0/79] Time 0.114 (0.114) Loss 0.2894 (0.2894) Prec 91.406%
(91.406%)
* Prec 88.930%
best acc: 90.330000
Epoch: [75][0/391] Time 0.184 (0.184) Data 0.147 (0.147) Loss
0.1086 (0.1086) Prec 96.094% (96.094%)
Epoch: [75][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.0748 (0.1063) Prec 98.438% (96.264%)
Epoch: [75][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0670 (0.1068) Prec 97.656% (96.191%)
Epoch: [75][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1138 (0.1071) Prec 94.531% (96.164%)
Validation starts
Test: [0/79] Time 0.116 (0.116) Loss 0.3772 (0.3772) Prec 89.062%
(89.062%)
* Prec 86.530%
best acc: 90.330000
Epoch: [76][0/391] Time 0.182 (0.182) Data 0.139 (0.139) Loss
0.1549 (0.1549) Prec 94.531% (94.531%)
Epoch: [76][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1110 (0.1075) Prec 95.312% (96.225%)
Epoch: [76][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.1959 (0.1074) Prec 93.750% (96.280%)
Epoch: [76][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1123 (0.1089) Prec 93.750% (96.177%)
Validation starts
Test: [0/79] Time 0.125 (0.125) Loss 0.2969 (0.2969) Prec 92.969%
(92.969%)
* Prec 88.380%
best acc: 90.330000
Epoch: [77][0/391] Time 0.187 (0.187) Data 0.139 (0.139) Loss
0.0746 (0.0746) Prec 97.656% (97.656%)
Epoch: [77][100/391] Time 0.055 (0.056) Data 0.001 (0.003) Loss
0.1289 (0.0952) Prec 94.531% (96.573%)
Epoch: [77][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2160 (0.0968) Prec 92.969% (96.506%)
Epoch: [77][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.1214 (0.1041) Prec 94.531% (96.307%)
Validation starts
Test: [0/79] Time 0.117 (0.117) Loss 0.3727 (0.3727) Prec 87.500%
(87.500%)
* Prec 88.120%
best acc: 90.330000
Epoch: [78][0/391] Time 0.179 (0.179) Data 0.138 (0.138) Loss
0.1156 (0.1156) Prec 96.094% (96.094%)

Epoch: [78][100/391] Time 0.055 (0.056) Data 0.001 (0.003) Loss
 0.0376 (0.1072) Prec 99.219% (96.179%)
 Epoch: [78][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.1094 (0.1039) Prec 95.312% (96.374%)
 Epoch: [78][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.1381 (0.1066) Prec 96.094% (96.260%)
 Validation starts
 Test: [0/79] Time 0.111 (0.111) Loss 0.3663 (0.3663) Prec 91.406%
 (91.406%)
 * Prec 89.440%
 best acc: 90.330000
 Epoch: [79][0/391] Time 0.159 (0.159) Data 0.120 (0.120) Loss
 0.1664 (0.1664) Prec 92.188% (92.188%)
 Epoch: [79][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.1016 (0.1041) Prec 96.094% (96.364%)
 Epoch: [79][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1488 (0.1051) Prec 94.531% (96.331%)
 Epoch: [79][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.1114 (0.1079) Prec 96.094% (96.190%)
 Validation starts
 Test: [0/79] Time 0.114 (0.114) Loss 0.3388 (0.3388) Prec 91.406%
 (91.406%)
 * Prec 89.160%
 best acc: 90.330000
 Epoch: [80][0/391] Time 0.198 (0.198) Data 0.153 (0.153) Loss
 0.0673 (0.0673) Prec 96.875% (96.875%)
 Epoch: [80][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.0343 (0.1037) Prec 99.219% (96.310%)
 Epoch: [80][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
 0.0820 (0.1048) Prec 97.656% (96.191%)
 Epoch: [80][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
 0.1093 (0.1099) Prec 96.094% (96.125%)
 Validation starts
 Test: [0/79] Time 0.126 (0.126) Loss 0.2510 (0.2510) Prec 94.531%
 (94.531%)
 * Prec 89.880%
 best acc: 90.330000
 Epoch: [81][0/391] Time 0.188 (0.188) Data 0.133 (0.133) Loss
 0.0559 (0.0559) Prec 97.656% (97.656%)
 Epoch: [81][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.0878 (0.1052) Prec 97.656% (96.326%)
 Epoch: [81][200/391] Time 0.055 (0.057) Data 0.002 (0.002) Loss
 0.1111 (0.1052) Prec 96.094% (96.331%)
 Epoch: [81][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.1491 (0.1070) Prec 94.531% (96.211%)
 Validation starts
 Test: [0/79] Time 0.108 (0.108) Loss 0.3946 (0.3946) Prec 92.969%
 (92.969%)

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* Prec 89.320%
best acc: 90.330000
Epoch: [82][0/391]      Time 0.173 (0.173)      Data 0.133 (0.133)      Loss
0.1145 (0.1145)      Prec 96.094% (96.094%)
Epoch: [82][100/391]    Time 0.055 (0.057)      Data 0.001 (0.003)      Loss
0.0958 (0.1031)      Prec 96.094% (96.364%)
Epoch: [82][200/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.1244 (0.1107)      Prec 96.875% (96.067%)
Epoch: [82][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.0805 (0.1118)      Prec 96.094% (96.021%)
Validation starts
Test: [0/79]      Time 0.115 (0.115)      Loss 0.2701 (0.2701)      Prec 90.625%
(90.625%)
* Prec 89.870%
best acc: 90.330000
Epoch: [83][0/391]      Time 0.196 (0.196)      Data 0.143 (0.143)      Loss
0.1152 (0.1152)      Prec 96.094% (96.094%)
Epoch: [83][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1083 (0.0988)      Prec 98.438% (96.566%)
Epoch: [83][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1498 (0.1015)      Prec 96.094% (96.432%)
Epoch: [83][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1483 (0.1039)      Prec 92.969% (96.320%)
Validation starts
Test: [0/79]      Time 0.107 (0.107)      Loss 0.2074 (0.2074)      Prec 92.188%
(92.188%)
* Prec 89.390%
best acc: 90.330000
Epoch: [84][0/391]      Time 0.192 (0.192)      Data 0.144 (0.144)      Loss
0.1509 (0.1509)      Prec 93.750% (93.750%)
Epoch: [84][100/391]    Time 0.055 (0.057)      Data 0.001 (0.003)      Loss
0.0726 (0.1044)      Prec 96.875% (96.388%)
Epoch: [84][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1597 (0.1086)      Prec 95.312% (96.168%)
Epoch: [84][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.0617 (0.1102)      Prec 98.438% (96.135%)
Validation starts
Test: [0/79]      Time 0.115 (0.115)      Loss 0.1860 (0.1860)      Prec 92.969%
(92.969%)
* Prec 87.920%
best acc: 90.330000
Epoch: [85][0/391]      Time 0.189 (0.189)      Data 0.137 (0.137)      Loss
0.0822 (0.0822)      Prec 96.875% (96.875%)
Epoch: [85][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.1109 (0.1072)      Prec 95.312% (96.047%)
Epoch: [85][200/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0879 (0.1082)      Prec 96.875% (96.016%)
Epoch: [85][300/391]    Time 0.055 (0.056)      Data 0.002 (0.002)      Loss

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0.0906 (0.1088) Prec 96.875% (96.070%)
Validation starts
Test: [0/79] Time 0.118 (0.118) Loss 0.2953 (0.2953) Prec 94.531% (94.531%)
* Prec 89.400%
best acc: 90.330000
Epoch: [86][0/391] Time 0.193 (0.193) Data 0.146 (0.146) Loss 0.1086 (0.1086) Prec 96.094% (96.094%)
Epoch: [86][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss 0.1273 (0.1018) Prec 96.875% (96.349%)
Epoch: [86][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss 0.2395 (0.1035) Prec 88.281% (96.343%)
Epoch: [86][300/391] Time 0.057 (0.056) Data 0.001 (0.002) Loss 0.1274 (0.1069) Prec 94.531% (96.218%)
Validation starts
Test: [0/79] Time 0.119 (0.119) Loss 0.4090 (0.4090) Prec 89.062% (89.062%)
* Prec 88.120%
best acc: 90.330000
Epoch: [87][0/391] Time 0.197 (0.197) Data 0.147 (0.147) Loss 0.1513 (0.1513) Prec 93.750% (93.750%)
Epoch: [87][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss 0.0548 (0.1092) Prec 98.438% (95.978%)
Epoch: [87][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss 0.0876 (0.1104) Prec 97.656% (96.000%)
Epoch: [87][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss 0.1445 (0.1123) Prec 96.875% (95.959%)
Validation starts
Test: [0/79] Time 0.124 (0.124) Loss 0.2758 (0.2758) Prec 93.750% (93.750%)
* Prec 88.910%
best acc: 90.330000
Epoch: [88][0/391] Time 0.182 (0.182) Data 0.141 (0.141) Loss 0.1809 (0.1809) Prec 93.750% (93.750%)
Epoch: [88][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss 0.1359 (0.0972) Prec 95.312% (96.566%)
Epoch: [88][200/391] Time 0.055 (0.057) Data 0.002 (0.002) Loss 0.1271 (0.1063) Prec 94.531% (96.296%)
Epoch: [88][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss 0.1072 (0.1074) Prec 96.875% (96.229%)
Validation starts
Test: [0/79] Time 0.134 (0.134) Loss 0.2931 (0.2931) Prec 91.406% (91.406%)
* Prec 88.610%
best acc: 90.330000
Epoch: [89][0/391] Time 0.183 (0.183) Data 0.134 (0.134) Loss 0.0952 (0.0952) Prec 94.531% (94.531%)
Epoch: [89][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss

0.0700 (0.1076) Prec 96.094% (96.187%)
Epoch: [89][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.0914 (0.1070) Prec 96.875% (96.206%)
Epoch: [89][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0518 (0.1096) Prec 98.438% (96.125%)
Validation starts
Test: [0/79] Time 0.130 (0.130) Loss 0.3402 (0.3402) Prec 91.406%
(91.406%)
* Prec 89.670%
best acc: 90.330000
Epoch: [90][0/391] Time 0.207 (0.207) Data 0.154 (0.154) Loss
0.1080 (0.1080) Prec 96.875% (96.875%)
Epoch: [90][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1141 (0.1040) Prec 96.875% (96.388%)
Epoch: [90][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.0485 (0.1089) Prec 97.656% (96.276%)
Epoch: [90][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1627 (0.1109) Prec 93.750% (96.156%)
Validation starts
Test: [0/79] Time 0.113 (0.113) Loss 0.2985 (0.2985) Prec 90.625%
(90.625%)
* Prec 89.330%
best acc: 90.330000
Epoch: [91][0/391] Time 0.190 (0.190) Data 0.134 (0.134) Loss
0.0558 (0.0558) Prec 98.438% (98.438%)
Epoch: [91][100/391] Time 0.055 (0.056) Data 0.001 (0.003) Loss
0.1117 (0.1026) Prec 96.875% (96.341%)
Epoch: [91][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1297 (0.1070) Prec 93.750% (96.168%)
Epoch: [91][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1509 (0.1085) Prec 94.531% (96.140%)
Validation starts
Test: [0/79] Time 0.107 (0.107) Loss 0.2575 (0.2575) Prec 92.969%
(92.969%)
* Prec 89.570%
best acc: 90.330000
Epoch: [92][0/391] Time 0.193 (0.193) Data 0.141 (0.141) Loss
0.0721 (0.0721) Prec 97.656% (97.656%)
Epoch: [92][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.0709 (0.0934) Prec 98.438% (96.852%)
Epoch: [92][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.0522 (0.0995) Prec 98.438% (96.595%)
Epoch: [92][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1548 (0.1059) Prec 94.531% (96.348%)
Validation starts
Test: [0/79] Time 0.115 (0.115) Loss 0.2892 (0.2892) Prec 91.406%
(91.406%)
* Prec 88.790%

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best acc: 90.330000
Epoch: [93][0/391]      Time 0.196 (0.196)      Data 0.146 (0.146)      Loss
0.1438 (0.1438)      Prec 93.750% (93.750%)
Epoch: [93][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1467 (0.1017)      Prec 93.750% (96.303%)
Epoch: [93][200/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1312 (0.1006)      Prec 94.531% (96.424%)
Epoch: [93][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.2172 (0.1064)      Prec 92.188% (96.179%)
Validation starts
Test: [0/79]      Time 0.111 (0.111)      Loss 0.3373 (0.3373)      Prec 89.844%
(89.844%)
* Prec 88.660%
best acc: 90.330000
Epoch: [94][0/391]      Time 0.162 (0.162)      Data 0.121 (0.121)      Loss
0.0197 (0.0197)      Prec 99.219% (99.219%)
Epoch: [94][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0727 (0.1041)      Prec 96.875% (96.511%)
Epoch: [94][200/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0750 (0.1081)      Prec 97.656% (96.284%)
Epoch: [94][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1543 (0.1085)      Prec 95.312% (96.221%)
Validation starts
Test: [0/79]      Time 0.116 (0.116)      Loss 0.3526 (0.3526)      Prec 90.625%
(90.625%)
* Prec 88.780%
best acc: 90.330000
Epoch: [95][0/391]      Time 0.213 (0.213)      Data 0.165 (0.165)      Loss
0.1785 (0.1785)      Prec 94.531% (94.531%)
Epoch: [95][100/391]    Time 0.055 (0.057)      Data 0.002 (0.003)      Loss
0.1064 (0.0976)      Prec 96.875% (96.558%)
Epoch: [95][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1408 (0.1063)      Prec 96.875% (96.218%)
Epoch: [95][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1105 (0.1056)      Prec 93.750% (96.273%)
Validation starts
Test: [0/79]      Time 0.115 (0.115)      Loss 0.4214 (0.4214)      Prec 88.281%
(88.281%)
* Prec 89.290%
best acc: 90.330000
Epoch: [96][0/391]      Time 0.166 (0.166)      Data 0.121 (0.121)      Loss
0.1106 (0.1106)      Prec 95.312% (95.312%)
Epoch: [96][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0745 (0.1003)      Prec 97.656% (96.488%)
Epoch: [96][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1001 (0.1088)      Prec 96.094% (96.171%)
Epoch: [96][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1326 (0.1086)      Prec 96.094% (96.174%)

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Validation starts
Test: [0/79]      Time 0.112 (0.112)      Loss 0.3720 (0.3720)      Prec 89.062%
(89.062%)
* Prec 88.900%
best acc: 90.330000
Epoch: [97][0/391]      Time 0.208 (0.208)      Data 0.152 (0.152)      Loss
0.0820 (0.0820)      Prec 96.875% (96.875%)
Epoch: [97][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.1202 (0.0997)      Prec 98.438% (96.573%)
Epoch: [97][200/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.1269 (0.1055)      Prec 93.750% (96.311%)
Epoch: [97][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.0729 (0.1089)      Prec 96.875% (96.159%)
Validation starts
Test: [0/79]      Time 0.105 (0.105)      Loss 0.2947 (0.2947)      Prec 89.844%
(89.844%)
* Prec 88.720%
best acc: 90.330000
Epoch: [98][0/391]      Time 0.173 (0.173)      Data 0.132 (0.132)      Loss
0.1311 (0.1311)      Prec 93.750% (93.750%)
Epoch: [98][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.1383 (0.1134)      Prec 94.531% (95.985%)
Epoch: [98][200/391]    Time 0.055 (0.057)      Data 0.001 (0.002)      Loss
0.0691 (0.1100)      Prec 97.656% (96.090%)
Epoch: [98][300/391]    Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.1211 (0.1102)      Prec 94.531% (96.174%)
Validation starts
Test: [0/79]      Time 0.114 (0.114)      Loss 0.4116 (0.4116)      Prec 90.625%
(90.625%)
* Prec 87.960%
best acc: 90.330000
Epoch: [99][0/391]      Time 0.169 (0.169)      Data 0.128 (0.128)      Loss
0.1000 (0.1000)      Prec 97.656% (97.656%)
Epoch: [99][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.0975 (0.1156)      Prec 96.094% (96.024%)
Epoch: [99][200/391]    Time 0.056 (0.057)      Data 0.002 (0.002)      Loss
0.1463 (0.1127)      Prec 96.094% (96.016%)
Epoch: [99][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.0833 (0.1130)      Prec 96.094% (96.044%)
Validation starts
Test: [0/79]      Time 0.106 (0.106)      Loss 0.1940 (0.1940)      Prec 95.312%
(95.312%)
* Prec 88.990%
best acc: 90.330000

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[27]: PATH = "result/pruningVGG16_quant/model_best.pth.tar"
checkpoint = torch.load(PATH)

```

```

model.load_state_dict(checkpoint['state_dict'])

model.cuda()

model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 8844/10000 (88%)

```

[28]: PATH = "result/VGG16_quant/model_best.pth.tar"
checkpoint = torch.load(PATH)
model = VGG16_quant()

model.load_state_dict(checkpoint['state_dict'])
model.cuda()
i = 0
for layer in model.modules():
    i = i+1
    if isinstance(layer, QuantConv2d):
        prune.ln_structured(layer, name="weight", amount=0.8, n = 1, dim = 0)

```

```

[29]: print(list(model.features[40].named_parameters())) # check whether there is
      ↪mask, weight_org, ...
print(model.features[40].weight) # check whether there are many zeros

```

```

[('act_alpha', Parameter containing:
tensor(0.6882, device='cuda:0', requires_grad=True)), ('weight_q', Parameter
containing:
tensor([[[[-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000],
          [-0.0000, -0.0000, -0.0000]],

```



```

[[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]],

...,

[[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]]],

[[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.5025, -0.0000],
  [ 0.0000,  0.5025, -0.0000]],

[[-0.5025, -0.5025, -0.0000],
  [-0.0000, -0.5025, -0.0000],
  [ 0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.5025]],

...,

[[[-0.5025, -0.5025, -0.0000],
  [ 1.0049, -0.5025, -0.5025],
  [-1.0049,  1.5074, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
  [ 1.5074,  0.0000, -0.0000],
  [-1.0049, -0.0000, -0.0000]],

[[-0.0000, -0.5025, -0.0000],
  [-0.5025, -0.5025, -0.0000],
  [ 0.0000, -0.0000, -0.0000]]],

```

```

[[[ 1.0049,  1.0049,  0.0000],
   [ 2.5123,  3.0147, -0.0000],
   [-0.0000,  0.5025, -0.0000]],

 [[ 1.0049,  0.5025, -0.5025],
   [ 0.5025,  0.5025, -0.0000],
   [ 0.0000, -0.0000, -0.0000]],

 [[-1.0049, -0.0000, -0.5025],
   [ 0.0000,  0.0000, -0.5025],
   [-0.0000, -0.5025, -0.0000]],

 ...,

 [[-2.0098, -2.0098, -1.0049],
   [-1.5074, -1.0049, -1.0049],
   [-0.0000, -0.0000, -0.0000]],

 [[-0.0000, -0.0000,  0.0000],
   [-1.0049,  2.5123,  0.0000],
   [-0.0000,  0.5025,  0.0000]],

 [[ 0.5025,  0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000],
   [ 0.0000, -0.0000, -0.0000]]],

 ...,

 [[[-0.0000, -0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000]],

 [[-0.0000, -0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000],
   [-0.0000,  0.0000, -0.0000]],

 [[-0.0000, -0.0000, -0.0000],
   [-0.0000,  0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000]],

 ...,

 [[-0.0000, -0.0000, -0.0000],
   [-0.0000, -0.0000, -0.0000],

```

```

[-0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000,  0.0000],
 [-0.0000, -0.0000,  0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000, -0.0000]]],

[[[-0.0000,  0.0000, -0.0000],
  [-0.0000,  0.0000, -0.0000],
  [ 0.5025, -0.5025, -0.0000]],

  [[-0.5025, -0.5025, -0.0000],
   [-0.0000, -1.0049, -0.5025],
   [ 0.5025,  0.5025, -0.0000]],

  [[ 0.0000, -0.0000, -0.0000],
   [ 1.0049, -0.5025, -0.5025],
   [-0.0000,  0.5025, -0.0000]],

  ...,

  [[-0.5025, -0.0000, -0.0000],
   [ 1.0049, -0.5025, -0.5025],
   [-0.5025,  0.5025, -0.0000]],

  [[-0.5025, -0.0000, -0.0000],
   [ 0.5025, -0.0000, -0.0000],
   [-1.0049, -0.5025, -0.0000]],

  [[-0.0000, -0.0000, -0.0000],
   [-0.5025, -0.0000, -0.0000],
   [ 0.0000, -0.0000, -0.0000]]],

[[[-0.0000, -1.0049, -0.5025],
  [ 0.5025,  2.0098, -1.0049],
  [ 3.5172,  0.0000, -0.0000]],

  [[-0.0000,  0.5025, -0.0000],
   [-0.5025,  3.0147, -1.5074],
   [ 2.0098,  3.5172,  0.0000]],

  [[-0.0000,  0.5025,  0.0000],
   [-0.5025, -1.0049, -0.0000],

```

```

[-0.5025,  3.5172,  0.0000]],

...,

[[-0.5025,  0.0000, -0.0000],
 [-3.5172, -3.5172, -1.0049],
 [ 3.5172,  3.5172, -0.0000]],

[[-1.0049, -0.5025,  0.0000],
 [-2.5123, -2.5123, -0.5025],
 [ 3.5172,  3.5172, -1.0049]],

[[ 0.5025,  0.5025,  0.0000],
 [ 1.0049,  0.5025, -0.0000],
 [ 1.0049, -0.0000, -0.0000]]], device='cuda:0', requires_grad=True)),
('weight_orig', Parameter containing:
tensor([[[[-1.7570e-05, -8.9569e-06,  1.0557e-06],
 [ 1.6296e-05,  8.2165e-05, -6.3749e-06],
 [ 6.6472e-05,  4.8311e-05, -2.4037e-06]],

[[-2.1359e-05,  4.6951e-05,  9.5514e-06],
 [-3.1503e-05,  2.1955e-05,  4.0229e-05],
 [ 3.5774e-05, -4.9366e-06, -1.1941e-05]],

[[ 9.7124e-06, -4.1532e-06, -1.4637e-05],
 [ 3.3825e-05, -4.3020e-05,  7.5637e-05],
 [ 3.3258e-05,  1.9391e-05,  2.0105e-05]],

...,

[[ 2.0706e-06, -4.3704e-06, -2.2272e-06],
 [ 1.1490e-05,  6.5538e-05, -1.2129e-05],
 [-2.7008e-05, -3.6061e-05, -2.4124e-05]],

[[ 3.5951e-05, -3.2756e-05,  9.0803e-06],
 [ 7.9857e-05,  3.4086e-05, -1.6635e-06],
 [ 9.4075e-06,  1.1999e-04, -4.0382e-05]],

[[ 1.8758e-05,  2.7689e-05,  1.7139e-05],
 [-4.1422e-05,  9.5847e-08,  8.1808e-06],
 [ 5.8145e-05, -1.9476e-05, -3.8287e-05]]],

[[[-1.5517e-04,  8.2275e-05, -1.4624e-04],
 [-7.3287e-04, -8.5354e-04, -6.3418e-04],
 [ 4.1919e-04,  2.4063e-03, -7.7785e-04]],

[[-1.2000e-03, -1.5387e-03, -1.7177e-04],

```

```

[-7.3789e-04, -1.6335e-03, -6.7902e-04],
[ 7.0406e-04,  2.3091e-05, -6.2861e-04]],

[[-1.2771e-04, -7.0946e-04, -1.3043e-04],
 [-1.4423e-04, -1.2029e-04, -7.1849e-04],
 [-8.7420e-05, -6.2121e-04, -9.8891e-04]],

...,

[[-1.2337e-03, -2.2048e-03, -3.9058e-04],
 [ 4.8492e-03, -2.3796e-03, -9.1923e-04],
 [-3.3027e-03,  5.4508e-03, -3.8683e-04]],

[[-7.4445e-05, -1.2986e-04, -6.3556e-05],
 [ 4.9579e-03,  1.7841e-04, -4.9176e-05],
 [-3.3219e-03, -2.3107e-05, -6.8390e-04]],

[[-7.4685e-04, -8.9520e-04, -7.8436e-05],
 [-1.7259e-03, -1.4290e-03, -4.8922e-04],
 [ 9.8038e-04, -4.4916e-05, -1.4341e-05]]],

[[[ 3.4294e-03,  3.8984e-03,  9.9814e-04],
 [ 9.7354e-03,  1.1741e-02, -3.5083e-04],
 [ 2.0531e-05,  2.5215e-03,  1.4772e-05]],

[[ 4.3132e-03,  1.9929e-03, -1.8510e-03],
 [ 1.7087e-03,  1.3521e-03, -1.9382e-04],
 [ 1.8923e-04, -3.1905e-04,  5.1719e-05]],

[[-3.2476e-03, -1.4931e-04, -1.8248e-03],
 [ 3.4393e-04,  7.2529e-04, -1.3105e-03],
 [-6.9917e-06, -1.0477e-03, -1.8952e-04]],

...,

[[-7.8411e-03, -8.1294e-03, -2.9235e-03],
 [-5.6326e-03, -4.1152e-03, -3.2639e-03],
 [-5.9923e-04, -2.5031e-04, -2.4815e-04]],

[[-4.0059e-04,  7.0557e-05,  8.7646e-04],
 [-3.5183e-03,  1.0080e-02,  7.0322e-04],
 [-4.8108e-04,  1.4617e-03,  5.6965e-04]],

[[ 2.8139e-03,  1.0220e-03, -1.0918e-04],
 [-6.5041e-04, -3.5622e-04,  4.9510e-05],
 [ 3.6596e-04,  1.3369e-04,  4.9963e-05]]],

```

...,

```
[[[-3.3910e-05, -5.3211e-05, -9.4828e-05],  
  [-2.0492e-04,  8.4314e-06, -5.3288e-05],  
  [-1.1428e-04,  1.5189e-04, -8.7737e-05]]],
```

```
[[[-8.5957e-05, -4.3550e-04, -3.7879e-04],  
  [-2.3909e-04, -2.5225e-04, -9.6679e-05],  
  [-5.6556e-05,  2.9708e-04,  5.0183e-05]]],
```

```
[[[-4.2509e-05, -1.2782e-04, -1.9607e-04],  
  [ 8.7912e-06,  3.6629e-04, -1.5277e-04],  
  [ 1.5602e-05, -1.6099e-04,  2.9055e-05]]],
```

...,

```
[[[-4.1510e-05, -3.0140e-04, -4.9867e-04],  
  [-1.5826e-04, -6.1948e-05, -6.0527e-05],  
  [-6.2447e-05,  2.8762e-04,  1.4419e-04]]],
```

```
[[[-2.4623e-04, -1.4424e-04, -2.9878e-05],  
  [-1.4807e-04,  3.3426e-04,  4.7028e-04],  
  [-2.8497e-04, -6.4177e-05,  6.2753e-04]]],
```

```
[[[-1.2092e-04, -2.5574e-04, -9.3159e-05],  
  [-1.9528e-04,  3.0568e-06,  6.1876e-06],  
  [-7.8125e-05,  2.7663e-04,  1.6424e-05]]],
```

```
[[[-9.5000e-05,  2.5334e-04, -6.2561e-05],  
  [-6.3878e-04,  7.6038e-04, -1.8044e-04],  
  [ 2.4336e-03, -1.2030e-03, -7.5862e-04]]],
```

```
[[[-8.8690e-04, -1.4285e-03, -2.4428e-04],  
  [-5.6330e-04, -4.3759e-03, -1.3184e-03],  
  [ 1.6374e-03,  1.2957e-03, -6.0395e-04]]],
```

```
[[ 1.7989e-04, -2.7715e-04,  6.6569e-05],  
  [ 3.1129e-03, -2.0130e-03, -1.0353e-03],  
  [-2.3236e-04,  1.3029e-03, -6.3600e-04]]],
```

...,

```
[[[-1.2368e-03, -7.4347e-04,  1.4939e-04],  
  [ 3.4365e-03, -1.1010e-03, -1.8069e-03],  
  [-2.0997e-03,  2.8714e-03, -2.9953e-04]]],
```

```

[[[-9.3919e-04, -6.7353e-04, -8.2398e-05],
 [ 2.4427e-03, -3.6102e-04, -1.3912e-04],
 [-4.4556e-03, -1.3877e-03, -6.9061e-04]],

 [[-4.8195e-04, -3.5676e-04, -6.6340e-05],
 [-1.6811e-03, -3.3772e-04, -9.5310e-05],
 [ 7.2166e-04, -7.6219e-06,  1.8374e-05]]],

 [[[-6.3399e-04, -2.8955e-03, -1.9088e-03],
 [ 2.2958e-03,  8.7109e-03, -2.9966e-03],
 [ 1.4137e-02,  6.4835e-04, -4.2956e-04]],

 [[-6.8038e-04,  2.1814e-03, -6.7176e-04],
 [-1.7984e-03,  1.1545e-02, -4.9462e-03],
 [ 7.3584e-03,  1.3596e-02,  4.4032e-04]],

 [[-7.2571e-04,  1.2464e-03,  4.3626e-04],
 [-1.9197e-03, -4.6240e-03, -5.5212e-04],
 [-1.2781e-03,  1.4136e-02,  3.7310e-04]],

 ...,

 [[-1.8318e-03,  5.1252e-04, -5.1797e-04],
 [-1.3777e-02, -1.3599e-02, -4.4352e-03],
 [ 1.4288e-02,  1.4019e-02, -2.4297e-04]],

 [[-2.8819e-03, -1.5895e-03,  3.7799e-04],
 [-9.1063e-03, -9.8070e-03, -1.6397e-03],
 [ 1.4011e-02,  1.4095e-02, -3.5573e-03]],

 [[ 1.2273e-03,  1.8809e-03,  1.7131e-04],
 [ 4.6441e-03,  2.6215e-03, -3.8282e-04],
 [ 4.0234e-03, -4.6813e-04, -1.0288e-04]]], device='cuda:0',
requires_grad=True)), ('weight_quant.wgt_alpha', Parameter containing:
tensor(3.5172, device='cuda:0', requires_grad=True))]
tensor([[[[-0.0000, -0.0000,  0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000]],

 [[-0.0000,  0.0000,  0.0000],
 [-0.0000,  0.0000,  0.0000],
 [ 0.0000, -0.0000, -0.0000]],

 [[ 0.0000, -0.0000, -0.0000],
 [ 0.0000, -0.0000,  0.0000],
 [ 0.0000,  0.0000,  0.0000]],

```

```

...,

[[ 0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000]],

[[ 0.0000, -0.0000,  0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000]],

[[ 0.0000,  0.0000,  0.0000],
 [-0.0000,  0.0000,  0.0000],
 [ 0.0000, -0.0000, -0.0000]],

[[[-0.0000,  0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000]],

...,

[[-0.0000, -0.0000, -0.0000],
 [ 0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [ 0.0000, -0.0000, -0.0000]],

[[[ 0.0000,  0.0000,  0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [ 0.0000,  0.0000,  0.0000]],

[[ 0.0000,  0.0000, -0.0000],

```



```

[ 0.0000,  0.0000, -0.0000],
[ 0.0000, -0.0000,  0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000]],

...,

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000]],

[[-0.0000,  0.0000,  0.0000],
 [-0.0000,  0.0000,  0.0000],
 [-0.0000,  0.0000,  0.0000]],

[[ 0.0000,  0.0000, -0.0000],
 [-0.0000, -0.0000,  0.0000],
 [ 0.0000,  0.0000,  0.0000]],

...,

[[[-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000, -0.0000],
 [-0.0000,  0.0000, -0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000,  0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [ 0.0000,  0.0000, -0.0000],
 [ 0.0000, -0.0000,  0.0000]],

...,

[[-0.0000, -0.0000, -0.0000],
 [-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000,  0.0000]],

[[-0.0000, -0.0000, -0.0000],
 [-0.0000,  0.0000,  0.0000],
 [-0.0000, -0.0000,  0.0000]],

[[-0.0000, -0.0000, -0.0000],
```

```

[-0.0000,  0.0000,  0.0000],
[-0.0000,  0.0000,  0.0000]]],

[[[-0.0000,  0.0000, -0.0000],
  [-0.0000,  0.0000, -0.0000],
  [ 0.0000, -0.0000, -0.0000]],

  [[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [ 0.0000,  0.0000, -0.0000]],

  [[ 0.0000, -0.0000,  0.0000],
  [ 0.0000, -0.0000, -0.0000],
  [-0.0000,  0.0000, -0.0000]],

  ...,

  [[-0.0000, -0.0000,  0.0000],
  [ 0.0000, -0.0000, -0.0000],
  [-0.0000,  0.0000, -0.0000]],

  [[-0.0000, -0.0000, -0.0000],
  [ 0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000]],

  [[-0.0000, -0.0000, -0.0000],
  [-0.0000, -0.0000, -0.0000],
  [ 0.0000, -0.0000,  0.0000]]],

  [[[-0.0006, -0.0029, -0.0019],
    [ 0.0023,  0.0087, -0.0030],
    [ 0.0141,  0.0006, -0.0004]],

    [[-0.0007,  0.0022, -0.0007],
    [-0.0018,  0.0115, -0.0049],
    [ 0.0074,  0.0136,  0.0004]],

    [[-0.0007,  0.0012,  0.0004],
    [-0.0019, -0.0046, -0.0006],
    [-0.0013,  0.0141,  0.0004]],

    ...,

    [[-0.0018,  0.0005, -0.0005],
    [-0.0138, -0.0136, -0.0044],
    [ 0.0143,  0.0140, -0.0002]],

```

```

[[[-0.0029, -0.0016,  0.0004],
  [-0.0091, -0.0098, -0.0016],
  [ 0.0140,  0.0141, -0.0036]],

 [[ 0.0012,  0.0019,  0.0002],
  [ 0.0046,  0.0026, -0.0004],
  [ 0.0040, -0.0005, -0.0001]]], device='cuda:0',
grad_fn=<MulBackward0>)

```

```

[30]: ### Check sparsity ###
mask1 = model.features[40].weight_mask
sparsity_mask1 = (mask1 == 0).sum() / mask1.nelement()

print("Sparsity level: ", sparsity_mask1)

```

Sparsity level: tensor(0.8008, device='cuda:0')

```

[31]: model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 1000/10000 (10%)

```

[32]: lr = 1e-1
weight_decay = 1e-4
epochs = 50
best_prec = 0

#model = nn.DataParallel(model).cuda()
model.cuda()
criterion = nn.CrossEntropyLoss().cuda()
optimizer = torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9,
    ↪weight_decay=weight_decay)

```

```

# cudnn.benchmark = True

if not os.path.exists('result'):
    os.makedirs('result')
fdir = 'result/' + 'pruning' + str(model_name)
if not os.path.exists(fdir):
    os.makedirs(fdir)

for epoch in range(0, epochs):
    adjust_learning_rate(optimizer, epoch)

    train(trainloader, model, criterion, optimizer, epoch)

    # evaluate on test set
    print("Validation starts")
    prec = validate(testloader, model, criterion)

    # remember best precision and save checkpoint
    is_best = prec > best_prec
    best_prec = max(prec, best_prec)
    print('best acc: {:.1f}'.format(best_prec))
    save_checkpoint({
        'epoch': epoch + 1,
        'state_dict': model.state_dict(),
        'best_prec': best_prec,
        'optimizer': optimizer.state_dict(),
    }, is_best, fdir)

```

```

Epoch: [0] [0/391]      Time 0.201 (0.201)      Data 0.164 (0.164)      Loss
5.1210 (5.1210)      Prec 9.375% (9.375%)
Epoch: [0] [100/391]    Time 0.053 (0.054)      Data 0.001 (0.003)      Loss
1.6083 (1.9917)      Prec 39.062% (26.021%)
Epoch: [0] [200/391]    Time 0.053 (0.053)      Data 0.002 (0.002)      Loss
1.4285 (1.7644)      Prec 45.312% (33.131%)
Epoch: [0] [300/391]    Time 0.053 (0.053)      Data 0.002 (0.002)      Loss
1.4696 (1.6411)      Prec 46.875% (37.843%)
Validation starts
Test: [0/79]      Time 0.120 (0.120)      Loss 1.2548 (1.2548)      Prec 56.250%
(56.250%)
* Prec 52.790%
best acc: 52.790000
Epoch: [1] [0/391]      Time 0.220 (0.220)      Data 0.164 (0.164)      Loss
1.2389 (1.2389)      Prec 53.125% (53.125%)
Epoch: [1] [100/391]    Time 0.052 (0.055)      Data 0.002 (0.003)      Loss
1.2399 (1.2558)      Prec 57.031% (54.053%)
Epoch: [1] [200/391]    Time 0.054 (0.054)      Data 0.002 (0.003)      Loss

```

0.9925 (1.2285) Prec 63.281% (55.480%)
Epoch: [1][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
1.2284 (1.2126) Prec 55.469% (56.289%)
Validation starts
Test: [0/79] Time 0.139 (0.139) Loss 1.1135 (1.1135) Prec 59.375%
(59.375%)
* Prec 57.920%
best acc: 57.920000
Epoch: [2][0/391] Time 0.195 (0.195) Data 0.154 (0.154) Loss
1.0613 (1.0613) Prec 58.594% (58.594%)
Epoch: [2][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
1.2231 (1.0891) Prec 57.812% (60.922%)
Epoch: [2][200/391] Time 0.053 (0.054) Data 0.001 (0.003) Loss
1.0377 (1.0885) Prec 57.031% (61.210%)
Epoch: [2][300/391] Time 0.052 (0.054) Data 0.002 (0.002) Loss
0.9455 (1.0799) Prec 71.875% (61.345%)
Validation starts
Test: [0/79] Time 0.139 (0.139) Loss 1.1237 (1.1237) Prec 61.719%
(61.719%)
* Prec 60.730%
best acc: 60.730000
Epoch: [3][0/391] Time 0.202 (0.202) Data 0.154 (0.154) Loss
1.1617 (1.1617) Prec 56.250% (56.250%)
Epoch: [3][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.9580 (1.0491) Prec 65.625% (62.376%)
Epoch: [3][200/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
1.2102 (1.0427) Prec 55.469% (62.551%)
Epoch: [3][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
1.0168 (1.0338) Prec 61.719% (62.978%)
Validation starts
Test: [0/79] Time 0.115 (0.115) Loss 1.1031 (1.1031) Prec 56.250%
(56.250%)
* Prec 59.070%
best acc: 60.730000
Epoch: [4][0/391] Time 0.207 (0.207) Data 0.152 (0.152) Loss
1.1651 (1.1651) Prec 60.156% (60.156%)
Epoch: [4][100/391] Time 0.053 (0.054) Data 0.002 (0.003) Loss
1.1758 (0.9962) Prec 62.500% (64.588%)
Epoch: [4][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
1.1037 (1.0034) Prec 60.938% (64.257%)
Epoch: [4][300/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
1.1574 (1.0078) Prec 57.031% (64.213%)
Validation starts
Test: [0/79] Time 0.116 (0.116) Loss 0.9847 (0.9847) Prec 67.188%
(67.188%)
* Prec 64.150%
best acc: 64.150000
Epoch: [5][0/391] Time 0.172 (0.172) Data 0.125 (0.125) Loss

```

0.8061 (0.8061)    Prec 68.750% (68.750%)
Epoch: [5][100/391]    Time 0.053 (0.054)    Data 0.001 (0.003)    Loss
0.8339 (0.9758)    Prec 73.438% (65.138%)
Epoch: [5][200/391]    Time 0.053 (0.053)    Data 0.001 (0.002)    Loss
0.9682 (0.9973)    Prec 63.281% (64.234%)
Epoch: [5][300/391]    Time 0.054 (0.053)    Data 0.001 (0.002)    Loss
0.9601 (0.9982)    Prec 64.062% (64.281%)
Validation starts
Test: [0/79]    Time 0.115 (0.115)    Loss 0.9558 (0.9558)    Prec 66.406%
(66.406%)
* Prec 64.690%
best acc: 64.690000
Epoch: [6][0/391]    Time 0.218 (0.218)    Data 0.166 (0.166)    Loss
1.3002 (1.3002)    Prec 51.562% (51.562%)
Epoch: [6][100/391]    Time 0.052 (0.055)    Data 0.002 (0.003)    Loss
1.0398 (0.9397)    Prec 66.406% (66.692%)
Epoch: [6][200/391]    Time 0.053 (0.054)    Data 0.002 (0.003)    Loss
0.9876 (0.9537)    Prec 64.844% (66.317%)
Epoch: [6][300/391]    Time 0.055 (0.054)    Data 0.001 (0.002)    Loss
0.9254 (0.9492)    Prec 67.188% (66.370%)
Validation starts
Test: [0/79]    Time 0.121 (0.121)    Loss 0.9917 (0.9917)    Prec 65.625%
(65.625%)
* Prec 63.960%
best acc: 64.690000
Epoch: [7][0/391]    Time 0.242 (0.242)    Data 0.194 (0.194)    Loss
1.0986 (1.0986)    Prec 60.156% (60.156%)
Epoch: [7][100/391]    Time 0.054 (0.056)    Data 0.002 (0.004)    Loss
1.0335 (0.9277)    Prec 64.844% (67.064%)
Epoch: [7][200/391]    Time 0.053 (0.055)    Data 0.002 (0.003)    Loss
0.9444 (0.9265)    Prec 67.188% (67.075%)
Epoch: [7][300/391]    Time 0.052 (0.054)    Data 0.002 (0.003)    Loss
0.8107 (0.9246)    Prec 72.656% (67.058%)
Validation starts
Test: [0/79]    Time 0.122 (0.122)    Loss 1.2427 (1.2427)    Prec 60.156%
(60.156%)
* Prec 59.630%
best acc: 64.690000
Epoch: [8][0/391]    Time 0.190 (0.190)    Data 0.148 (0.148)    Loss
0.8375 (0.8375)    Prec 71.875% (71.875%)
Epoch: [8][100/391]    Time 0.053 (0.054)    Data 0.002 (0.003)    Loss
1.1432 (0.9215)    Prec 60.156% (66.979%)
Epoch: [8][200/391]    Time 0.053 (0.054)    Data 0.001 (0.002)    Loss
0.8125 (0.9155)    Prec 74.219% (67.250%)
Epoch: [8][300/391]    Time 0.053 (0.053)    Data 0.002 (0.002)    Loss
0.8569 (0.9067)    Prec 68.750% (67.590%)
Validation starts
Test: [0/79]    Time 0.127 (0.127)    Loss 0.9333 (0.9333)    Prec 64.844%

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(64.844%)
* Prec 66.900%
best acc: 66.900000
Epoch: [9][0/391]      Time 0.229 (0.229)      Data 0.182 (0.182)      Loss
1.0246 (1.0246)      Prec 66.406% (66.406%)
Epoch: [9][100/391]    Time 0.055 (0.056)      Data 0.002 (0.004)      Loss
0.8977 (0.8927)      Prec 67.969% (68.634%)
Epoch: [9][200/391]    Time 0.052 (0.055)      Data 0.001 (0.003)      Loss
0.9931 (0.8956)      Prec 64.062% (68.462%)
Epoch: [9][300/391]    Time 0.053 (0.054)      Data 0.002 (0.002)      Loss
0.8295 (0.8918)      Prec 71.094% (68.579%)
Validation starts
Test: [0/79]      Time 0.129 (0.129)      Loss 1.0641 (1.0641)      Prec 62.500%
(62.500%)
* Prec 59.960%
best acc: 66.900000
Epoch: [10][0/391]     Time 0.214 (0.214)      Data 0.159 (0.159)      Loss
0.8662 (0.8662)      Prec 65.625% (65.625%)
Epoch: [10][100/391]   Time 0.052 (0.055)      Data 0.001 (0.003)      Loss
0.7883 (0.8932)      Prec 73.438% (68.410%)
Epoch: [10][200/391]   Time 0.054 (0.054)      Data 0.001 (0.002)      Loss
0.8697 (0.8937)      Prec 69.531% (68.284%)
Epoch: [10][300/391]   Time 0.054 (0.054)      Data 0.014 (0.002)      Loss
0.9198 (0.8833)      Prec 67.188% (68.529%)
Validation starts
Test: [0/79]      Time 0.115 (0.115)      Loss 0.9270 (0.9270)      Prec 67.969%
(67.969%)
* Prec 62.470%
best acc: 66.900000
Epoch: [11][0/391]     Time 0.194 (0.194)      Data 0.152 (0.152)      Loss
0.8839 (0.8839)      Prec 64.844% (64.844%)
Epoch: [11][100/391]   Time 0.052 (0.055)      Data 0.001 (0.003)      Loss
1.0120 (0.8796)      Prec 67.188% (69.036%)
Epoch: [11][200/391]   Time 0.052 (0.054)      Data 0.001 (0.002)      Loss
0.8324 (0.8769)      Prec 67.969% (68.964%)
Epoch: [11][300/391]   Time 0.053 (0.054)      Data 0.001 (0.002)      Loss
0.9781 (0.8748)      Prec 58.594% (69.002%)
Validation starts
Test: [0/79]      Time 0.127 (0.127)      Loss 0.9067 (0.9067)      Prec 70.312%
(70.312%)
* Prec 68.760%
best acc: 68.760000
Epoch: [12][0/391]     Time 0.210 (0.210)      Data 0.162 (0.162)      Loss
0.9172 (0.9172)      Prec 64.844% (64.844%)
Epoch: [12][100/391]   Time 0.054 (0.056)      Data 0.001 (0.003)      Loss
0.9412 (0.8538)      Prec 68.750% (69.624%)
Epoch: [12][200/391]   Time 0.054 (0.055)      Data 0.002 (0.003)      Loss
0.8519 (0.8534)      Prec 75.781% (69.485%)

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Epoch: [12][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss 0.8161 (0.8509) Prec 68.750% (69.705%)
Validation starts
Test: [0/79] Time 0.142 (0.142) Loss 0.8598 (0.8598) Prec 67.969% (67.969%)
* Prec 66.520%
best acc: 68.760000

Epoch: [13][0/391] Time 0.190 (0.190) Data 0.148 (0.148) Loss 0.7879 (0.7879) Prec 69.531% (69.531%)
Epoch: [13][100/391] Time 0.053 (0.054) Data 0.001 (0.003) Loss 0.8234 (0.8347) Prec 71.875% (70.057%)
Epoch: [13][200/391] Time 0.052 (0.053) Data 0.001 (0.002) Loss 0.8596 (0.8400) Prec 68.750% (70.103%)
Epoch: [13][300/391] Time 0.053 (0.053) Data 0.002 (0.002) Loss 0.9083 (0.8417) Prec 70.312% (70.079%)
Validation starts
Test: [0/79] Time 0.119 (0.119) Loss 0.8752 (0.8752) Prec 71.094% (71.094%)
* Prec 68.990%
best acc: 68.990000

Epoch: [14][0/391] Time 0.225 (0.225) Data 0.170 (0.170) Loss 0.8435 (0.8435) Prec 71.875% (71.875%)
Epoch: [14][100/391] Time 0.054 (0.055) Data 0.002 (0.004) Loss 0.7869 (0.8403) Prec 68.750% (70.104%)
Epoch: [14][200/391] Time 0.052 (0.054) Data 0.001 (0.003) Loss 0.7748 (0.8300) Prec 75.000% (70.487%)
Epoch: [14][300/391] Time 0.052 (0.054) Data 0.002 (0.002) Loss 0.6680 (0.8310) Prec 78.125% (70.486%)
Validation starts
Test: [0/79] Time 0.127 (0.127) Loss 0.8170 (0.8170) Prec 69.531% (69.531%)
* Prec 71.530%
best acc: 71.530000

Epoch: [15][0/391] Time 0.211 (0.211) Data 0.162 (0.162) Loss 0.8011 (0.8011) Prec 71.094% (71.094%)
Epoch: [15][100/391] Time 0.054 (0.055) Data 0.001 (0.003) Loss 0.9025 (0.8362) Prec 67.969% (70.282%)
Epoch: [15][200/391] Time 0.053 (0.054) Data 0.001 (0.003) Loss 0.7886 (0.8301) Prec 72.656% (70.849%)
Epoch: [15][300/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss 0.8107 (0.8312) Prec 68.750% (70.764%)
Validation starts
Test: [0/79] Time 0.146 (0.146) Loss 0.9282 (0.9282) Prec 69.531% (69.531%)
* Prec 68.200%
best acc: 71.530000

Epoch: [16][0/391] Time 0.238 (0.238) Data 0.197 (0.197) Loss 0.9162 (0.9162) Prec 73.438% (73.438%)

Epoch: [16][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
 0.8425 (0.8319) Prec 68.750% (70.792%)
 Epoch: [16][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
 0.8180 (0.8231) Prec 71.094% (70.965%)
 Epoch: [16][300/391] Time 0.054 (0.054) Data 0.001 (0.002) Loss
 0.7103 (0.8192) Prec 75.781% (71.156%)
 Validation starts
 Test: [0/79] Time 0.125 (0.125) Loss 0.8542 (0.8542) Prec 70.312%
 (70.312%)
 * Prec 67.120%
 best acc: 71.530000
 Epoch: [17][0/391] Time 0.200 (0.200) Data 0.159 (0.159) Loss
 0.8212 (0.8212) Prec 75.000% (75.000%)
 Epoch: [17][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
 0.7956 (0.8215) Prec 72.656% (70.622%)
 Epoch: [17][200/391] Time 0.053 (0.054) Data 0.002 (0.003) Loss
 0.7826 (0.8099) Prec 70.312% (71.206%)
 Epoch: [17][300/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
 0.7249 (0.8118) Prec 73.438% (71.309%)
 Validation starts
 Test: [0/79] Time 0.134 (0.134) Loss 0.7933 (0.7933) Prec 71.875%
 (71.875%)
 * Prec 71.390%
 best acc: 71.530000
 Epoch: [18][0/391] Time 0.190 (0.190) Data 0.137 (0.137) Loss
 0.9350 (0.9350) Prec 65.625% (65.625%)
 Epoch: [18][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
 0.8688 (0.8235) Prec 71.094% (70.792%)
 Epoch: [18][200/391] Time 0.053 (0.054) Data 0.002 (0.003) Loss
 0.7441 (0.8111) Prec 71.875% (71.164%)
 Epoch: [18][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
 0.8987 (0.8183) Prec 65.625% (70.956%)
 Validation starts
 Test: [0/79] Time 0.142 (0.142) Loss 0.9181 (0.9181) Prec 70.312%
 (70.312%)
 * Prec 68.710%
 best acc: 71.530000
 Epoch: [19][0/391] Time 0.209 (0.209) Data 0.155 (0.155) Loss
 0.8636 (0.8636) Prec 72.656% (72.656%)
 Epoch: [19][100/391] Time 0.054 (0.055) Data 0.001 (0.003) Loss
 0.7548 (0.8066) Prec 75.781% (71.187%)
 Epoch: [19][200/391] Time 0.052 (0.054) Data 0.001 (0.002) Loss
 0.7946 (0.8115) Prec 73.438% (71.090%)
 Epoch: [19][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
 0.7275 (0.8056) Prec 71.094% (71.475%)
 Validation starts
 Test: [0/79] Time 0.116 (0.116) Loss 0.8601 (0.8601) Prec 75.000%
 (75.000%)

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* Prec 70.770%
best acc: 71.530000
Epoch: [20][0/391]      Time 0.210 (0.210)      Data 0.160 (0.160)      Loss
0.7675 (0.7675)      Prec 73.438% (73.438%)
Epoch: [20][100/391]    Time 0.053 (0.055)      Data 0.002 (0.004)      Loss
0.8019 (0.7980)      Prec 73.438% (71.921%)
Epoch: [20][200/391]    Time 0.053 (0.054)      Data 0.002 (0.003)      Loss
0.8439 (0.7941)      Prec 71.875% (72.128%)
Epoch: [20][300/391]    Time 0.053 (0.054)      Data 0.002 (0.002)      Loss
0.7257 (0.7978)      Prec 78.125% (72.023%)
Validation starts
Test: [0/79]      Time 0.141 (0.141)      Loss 0.7811 (0.7811)      Prec 75.000%
(75.000%)
* Prec 71.260%
best acc: 71.530000
Epoch: [21][0/391]      Time 0.209 (0.209)      Data 0.161 (0.161)      Loss
0.9087 (0.9087)      Prec 67.188% (67.188%)
Epoch: [21][100/391]    Time 0.052 (0.055)      Data 0.001 (0.003)      Loss
0.6873 (0.8155)      Prec 77.344% (71.163%)
Epoch: [21][200/391]    Time 0.053 (0.054)      Data 0.001 (0.002)      Loss
0.9024 (0.8003)      Prec 66.406% (71.949%)
Epoch: [21][300/391]    Time 0.053 (0.054)      Data 0.001 (0.002)      Loss
0.9214 (0.8002)      Prec 66.406% (71.994%)
Validation starts
Test: [0/79]      Time 0.119 (0.119)      Loss 0.7971 (0.7971)      Prec 75.000%
(75.000%)
* Prec 66.490%
best acc: 71.530000
Epoch: [22][0/391]      Time 0.182 (0.182)      Data 0.143 (0.143)      Loss
0.7338 (0.7338)      Prec 75.000% (75.000%)
Epoch: [22][100/391]    Time 0.053 (0.054)      Data 0.001 (0.003)      Loss
0.8950 (0.8013)      Prec 70.312% (71.713%)
Epoch: [22][200/391]    Time 0.053 (0.054)      Data 0.001 (0.002)      Loss
0.9733 (0.8093)      Prec 66.406% (71.537%)
Epoch: [22][300/391]    Time 0.053 (0.053)      Data 0.002 (0.002)      Loss
0.7030 (0.8002)      Prec 78.125% (71.898%)
Validation starts
Test: [0/79]      Time 0.112 (0.112)      Loss 0.8020 (0.8020)      Prec 71.875%
(71.875%)
* Prec 71.520%
best acc: 71.530000
Epoch: [23][0/391]      Time 0.193 (0.193)      Data 0.143 (0.143)      Loss
0.8598 (0.8598)      Prec 71.094% (71.094%)
Epoch: [23][100/391]    Time 0.055 (0.055)      Data 0.002 (0.003)      Loss
0.7946 (0.7876)      Prec 71.094% (72.447%)
Epoch: [23][200/391]    Time 0.055 (0.054)      Data 0.002 (0.003)      Loss
0.7734 (0.7959)      Prec 71.875% (71.898%)
Epoch: [23][300/391]    Time 0.055 (0.054)      Data 0.002 (0.002)      Loss

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0.6862 (0.8109) Prec 72.656% (71.384%)
Validation starts
Test: [0/79] Time 0.132 (0.132) Loss 0.7940 (0.7940) Prec 73.438%
(73.438%)
* Prec 73.150%
best acc: 73.150000
Epoch: [24][0/391] Time 0.191 (0.191) Data 0.152 (0.152) Loss
0.7228 (0.7228) Prec 72.656% (72.656%)
Epoch: [24][100/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.7385 (0.7923) Prec 76.562% (72.123%)
Epoch: [24][200/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.6829 (0.7902) Prec 75.000% (72.100%)
Epoch: [24][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.6279 (0.7921) Prec 81.250% (72.114%)
Validation starts
Test: [0/79] Time 0.137 (0.137) Loss 0.7046 (0.7046) Prec 75.781%
(75.781%)
* Prec 70.680%
best acc: 73.150000
Epoch: [25][0/391] Time 0.199 (0.199) Data 0.155 (0.155) Loss
0.6621 (0.6621) Prec 76.562% (76.562%)
Epoch: [25][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.7467 (0.7844) Prec 75.000% (72.378%)
Epoch: [25][200/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.8593 (0.7830) Prec 67.969% (72.629%)
Epoch: [25][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.6924 (0.7851) Prec 71.094% (72.399%)
Validation starts
Test: [0/79] Time 0.121 (0.121) Loss 0.9531 (0.9531) Prec 67.188%
(67.188%)
* Prec 67.330%
best acc: 73.150000
Epoch: [26][0/391] Time 0.197 (0.197) Data 0.151 (0.151) Loss
0.8157 (0.8157) Prec 68.750% (68.750%)
Epoch: [26][100/391] Time 0.054 (0.055) Data 0.001 (0.003) Loss
0.7350 (0.7839) Prec 75.000% (72.703%)
Epoch: [26][200/391] Time 0.054 (0.054) Data 0.002 (0.003) Loss
0.7772 (0.7884) Prec 72.656% (72.233%)
Epoch: [26][300/391] Time 0.052 (0.054) Data 0.002 (0.002) Loss
0.8903 (0.7898) Prec 67.188% (72.096%)
Validation starts
Test: [0/79] Time 0.139 (0.139) Loss 0.7883 (0.7883) Prec 75.000%
(75.000%)
* Prec 70.610%
best acc: 73.150000
Epoch: [27][0/391] Time 0.182 (0.182) Data 0.135 (0.135) Loss
0.8233 (0.8233) Prec 70.312% (70.312%)
Epoch: [27][100/391] Time 0.054 (0.054) Data 0.001 (0.003) Loss

0.7926 (0.8132) Prec 68.750% (72.061%)
Epoch: [27][200/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.9735 (0.7957) Prec 62.500% (72.306%)
Epoch: [27][300/391] Time 0.054 (0.054) Data 0.001 (0.002) Loss
0.7574 (0.7928) Prec 73.438% (72.233%)
Validation starts
Test: [0/79] Time 0.120 (0.120) Loss 0.8192 (0.8192) Prec 71.875%
(71.875%)
* Prec 67.570%
best acc: 73.150000
Epoch: [28][0/391] Time 0.208 (0.208) Data 0.155 (0.155) Loss
0.8667 (0.8667) Prec 69.531% (69.531%)
Epoch: [28][100/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.9356 (0.7885) Prec 64.062% (72.045%)
Epoch: [28][200/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.7874 (0.7824) Prec 72.656% (72.423%)
Epoch: [28][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.7826 (0.7824) Prec 71.094% (72.433%)
Validation starts
Test: [0/79] Time 0.134 (0.134) Loss 0.7995 (0.7995) Prec 77.344%
(77.344%)
* Prec 72.190%
best acc: 73.150000
Epoch: [29][0/391] Time 0.191 (0.191) Data 0.150 (0.150) Loss
0.7554 (0.7554) Prec 78.906% (78.906%)
Epoch: [29][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
0.7535 (0.7654) Prec 71.875% (73.097%)
Epoch: [29][200/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.7745 (0.7757) Prec 72.656% (72.637%)
Epoch: [29][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.7136 (0.7813) Prec 75.000% (72.516%)
Validation starts
Test: [0/79] Time 0.150 (0.150) Loss 0.7806 (0.7806) Prec 71.875%
(71.875%)
* Prec 71.020%
best acc: 73.150000
Epoch: [30][0/391] Time 0.196 (0.196) Data 0.149 (0.149) Loss
0.6940 (0.6940) Prec 75.000% (75.000%)
Epoch: [30][100/391] Time 0.053 (0.054) Data 0.001 (0.003) Loss
0.7314 (0.7691) Prec 76.562% (73.113%)
Epoch: [30][200/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.6180 (0.7793) Prec 78.125% (72.606%)
Epoch: [30][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.9464 (0.7774) Prec 66.406% (72.646%)
Validation starts
Test: [0/79] Time 0.123 (0.123) Loss 0.9837 (0.9837) Prec 66.406%
(66.406%)
* Prec 68.390%

```

best acc: 73.150000
Epoch: [31][0/391]      Time 0.216 (0.216)      Data 0.167 (0.167)      Loss
0.9229 (0.9229)      Prec 64.844% (64.844%)
Epoch: [31][100/391]    Time 0.053 (0.055)      Data 0.002 (0.003)      Loss
0.8799 (0.7618)      Prec 67.188% (73.159%)
Epoch: [31][200/391]    Time 0.053 (0.054)      Data 0.002 (0.003)      Loss
0.7357 (0.7648)      Prec 78.906% (72.994%)
Epoch: [31][300/391]    Time 0.054 (0.054)      Data 0.002 (0.002)      Loss
0.8221 (0.7758)      Prec 71.875% (72.623%)
Validation starts
Test: [0/79]      Time 0.152 (0.152)      Loss 0.8413 (0.8413)      Prec 74.219%
(74.219%)
* Prec 68.890%
best acc: 73.150000
Epoch: [32][0/391]      Time 0.239 (0.239)      Data 0.192 (0.192)      Loss
0.7822 (0.7822)      Prec 74.219% (74.219%)
Epoch: [32][100/391]    Time 0.054 (0.056)      Data 0.002 (0.004)      Loss
0.8787 (0.7719)      Prec 68.750% (73.120%)
Epoch: [32][200/391]    Time 0.053 (0.055)      Data 0.002 (0.003)      Loss
0.9211 (0.7763)      Prec 67.969% (72.932%)
Epoch: [32][300/391]    Time 0.054 (0.054)      Data 0.002 (0.002)      Loss
0.8324 (0.7820)      Prec 70.312% (72.617%)
Validation starts
Test: [0/79]      Time 0.140 (0.140)      Loss 0.9163 (0.9163)      Prec 69.531%
(69.531%)
* Prec 63.950%
best acc: 73.150000
Epoch: [33][0/391]      Time 0.217 (0.217)      Data 0.169 (0.169)      Loss
0.5774 (0.5774)      Prec 82.031% (82.031%)
Epoch: [33][100/391]    Time 0.053 (0.055)      Data 0.001 (0.003)      Loss
0.8482 (0.7735)      Prec 67.188% (72.857%)
Epoch: [33][200/391]    Time 0.053 (0.054)      Data 0.002 (0.003)      Loss
0.8563 (0.7773)      Prec 67.969% (72.660%)
Epoch: [33][300/391]    Time 0.054 (0.054)      Data 0.002 (0.002)      Loss
0.6577 (0.7743)      Prec 75.781% (72.721%)
Validation starts
Test: [0/79]      Time 0.135 (0.135)      Loss 0.7242 (0.7242)      Prec 75.781%
(75.781%)
* Prec 73.270%
best acc: 73.270000
Epoch: [34][0/391]      Time 0.175 (0.175)      Data 0.128 (0.128)      Loss
0.7956 (0.7956)      Prec 70.312% (70.312%)
Epoch: [34][100/391]    Time 0.053 (0.054)      Data 0.001 (0.003)      Loss
0.6601 (0.7764)      Prec 78.906% (72.540%)
Epoch: [34][200/391]    Time 0.053 (0.054)      Data 0.001 (0.002)      Loss
0.8731 (0.7781)      Prec 67.188% (72.695%)
Epoch: [34][300/391]    Time 0.053 (0.053)      Data 0.001 (0.002)      Loss
0.7216 (0.7752)      Prec 74.219% (72.937%)

```

Validation starts

Test: [0/79] Time 0.122 (0.122) Loss 0.7788 (0.7788) Prec 71.875%
(71.875%)

* Prec 70.370%

best acc: 73.270000

Epoch: [35][0/391] Time 0.226 (0.226) Data 0.176 (0.176) Loss
0.8295 (0.8295) Prec 71.875% (71.875%)

Epoch: [35][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.9034 (0.7601) Prec 63.281% (73.438%)

Epoch: [35][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.8590 (0.7687) Prec 70.312% (73.045%)

Epoch: [35][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7476 (0.7739) Prec 71.875% (72.937%)

Validation starts

Test: [0/79] Time 0.121 (0.121) Loss 0.9376 (0.9376) Prec 68.750%
(68.750%)

* Prec 69.300%

best acc: 73.270000

Epoch: [36][0/391] Time 0.195 (0.195) Data 0.147 (0.147) Loss
0.9521 (0.9521) Prec 70.312% (70.312%)

Epoch: [36][100/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.6036 (0.7734) Prec 77.344% (73.058%)

Epoch: [36][200/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.6826 (0.7648) Prec 75.781% (73.228%)

Epoch: [36][300/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.7669 (0.7697) Prec 75.000% (72.963%)

Validation starts

Test: [0/79] Time 0.133 (0.133) Loss 0.8860 (0.8860) Prec 70.312%
(70.312%)

* Prec 71.250%

best acc: 73.270000

Epoch: [37][0/391] Time 0.208 (0.208) Data 0.153 (0.153) Loss
0.8462 (0.8462) Prec 71.875% (71.875%)

Epoch: [37][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.7323 (0.7687) Prec 71.875% (73.043%)

Epoch: [37][200/391] Time 0.052 (0.054) Data 0.002 (0.002) Loss
0.6910 (0.7649) Prec 75.781% (73.169%)

Epoch: [37][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7487 (0.7621) Prec 71.875% (73.349%)

Validation starts

Test: [0/79] Time 0.124 (0.124) Loss 0.8383 (0.8383) Prec 71.094%
(71.094%)

* Prec 72.690%

best acc: 73.270000

Epoch: [38][0/391] Time 0.206 (0.206) Data 0.159 (0.159) Loss
0.6767 (0.6767) Prec 78.906% (78.906%)

Epoch: [38][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.6908 (0.7600) Prec 75.781% (73.329%)

Epoch: [38][200/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.6616 (0.7648) Prec 82.031% (73.142%)

Epoch: [38][300/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.6939 (0.7585) Prec 77.344% (73.427%)

Validation starts

Test: [0/79] Time 0.125 (0.125) Loss 0.7172 (0.7172) Prec 76.562%
(76.562%)

* Prec 73.010%

best acc: 73.270000

Epoch: [39][0/391] Time 0.212 (0.212) Data 0.161 (0.161) Loss
0.7138 (0.7138) Prec 75.781% (75.781%)

Epoch: [39][100/391] Time 0.054 (0.055) Data 0.001 (0.003) Loss
0.9237 (0.7650) Prec 68.750% (73.236%)

Epoch: [39][200/391] Time 0.046 (0.054) Data 0.002 (0.002) Loss
0.6702 (0.7564) Prec 78.906% (73.585%)

Epoch: [39][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7746 (0.7568) Prec 68.750% (73.596%)

Validation starts

Test: [0/79] Time 0.124 (0.124) Loss 0.6932 (0.6932) Prec 77.344%
(77.344%)

* Prec 73.580%

best acc: 73.580000

Epoch: [40][0/391] Time 0.195 (0.195) Data 0.149 (0.149) Loss
0.8297 (0.8297) Prec 69.531% (69.531%)

Epoch: [40][100/391] Time 0.052 (0.054) Data 0.001 (0.003) Loss
0.6491 (0.7434) Prec 75.000% (73.917%)

Epoch: [40][200/391] Time 0.054 (0.054) Data 0.001 (0.002) Loss
0.7299 (0.7537) Prec 74.219% (73.577%)

Epoch: [40][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.6793 (0.7579) Prec 73.438% (73.430%)

Validation starts

Test: [0/79] Time 0.121 (0.121) Loss 0.8217 (0.8217) Prec 74.219%
(74.219%)

* Prec 71.660%

best acc: 73.580000

Epoch: [41][0/391] Time 0.210 (0.210) Data 0.162 (0.162) Loss
0.5837 (0.5837) Prec 80.469% (80.469%)

Epoch: [41][100/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.6707 (0.7532) Prec 73.438% (73.515%)

Epoch: [41][200/391] Time 0.054 (0.054) Data 0.001 (0.003) Loss
0.7570 (0.7592) Prec 71.094% (73.274%)

Epoch: [41][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.9236 (0.7584) Prec 67.969% (73.282%)

Validation starts

Test: [0/79] Time 0.140 (0.140) Loss 0.7886 (0.7886) Prec 75.781%
(75.781%)

* Prec 72.200%

best acc: 73.580000

Epoch: [42][0/391] Time 0.193 (0.193) Data 0.150 (0.150) Loss
0.7864 (0.7864) Prec 71.875% (71.875%)

Epoch: [42][100/391] Time 0.052 (0.054) Data 0.001 (0.003) Loss
0.7830 (0.7367) Prec 69.531% (74.234%)

Epoch: [42][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7879 (0.7561) Prec 73.438% (73.558%)

Epoch: [42][300/391] Time 0.053 (0.053) Data 0.001 (0.002) Loss
0.7089 (0.7577) Prec 73.438% (73.559%)

Validation starts
Test: [0/79] Time 0.122 (0.122) Loss 0.7811 (0.7811) Prec 73.438%
(73.438%)
* Prec 71.930%
best acc: 73.580000

Epoch: [43][0/391] Time 0.198 (0.198) Data 0.154 (0.154) Loss
0.6001 (0.6001) Prec 81.250% (81.250%)

Epoch: [43][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
0.8091 (0.7503) Prec 71.094% (73.376%)

Epoch: [43][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.6972 (0.7549) Prec 72.656% (73.185%)

Epoch: [43][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7667 (0.7534) Prec 73.438% (73.412%)

Validation starts
Test: [0/79] Time 0.120 (0.120) Loss 0.7222 (0.7222) Prec 76.562%
(76.562%)
* Prec 72.240%
best acc: 73.580000

Epoch: [44][0/391] Time 0.209 (0.209) Data 0.170 (0.170) Loss
0.6734 (0.6734) Prec 73.438% (73.438%)

Epoch: [44][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.6714 (0.7601) Prec 76.562% (73.801%)

Epoch: [44][200/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.7405 (0.7534) Prec 72.656% (73.850%)

Epoch: [44][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.8105 (0.7523) Prec 72.656% (73.848%)

Validation starts
Test: [0/79] Time 0.122 (0.122) Loss 0.6629 (0.6629) Prec 79.688%
(79.688%)
* Prec 73.950%
best acc: 73.950000

Epoch: [45][0/391] Time 0.202 (0.202) Data 0.152 (0.152) Loss
0.8669 (0.8669) Prec 69.531% (69.531%)

Epoch: [45][100/391] Time 0.054 (0.055) Data 0.002 (0.003) Loss
0.5546 (0.7415) Prec 85.156% (73.708%)

Epoch: [45][200/391] Time 0.054 (0.054) Data 0.002 (0.003) Loss
0.6208 (0.7481) Prec 77.344% (73.461%)

Epoch: [45][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.7622 (0.7483) Prec 71.875% (73.510%)

Validation starts

Test: [0/79] Time 0.143 (0.143) Loss 0.7509 (0.7509) Prec 76.562%
(76.562%)

* Prec 72.770%

best acc: 73.950000

Epoch: [46][0/391] Time 0.209 (0.209) Data 0.171 (0.171) Loss
0.6596 (0.6596) Prec 74.219% (74.219%)

Epoch: [46][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
0.9535 (0.7347) Prec 65.625% (74.435%)

Epoch: [46][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.8736 (0.7413) Prec 71.094% (74.052%)

Epoch: [46][300/391] Time 0.054 (0.054) Data 0.001 (0.002) Loss
0.7681 (0.7447) Prec 72.656% (73.923%)

Validation starts

Test: [0/79] Time 0.134 (0.134) Loss 0.7436 (0.7436) Prec 74.219%
(74.219%)

* Prec 73.840%

best acc: 73.950000

Epoch: [47][0/391] Time 0.207 (0.207) Data 0.159 (0.159) Loss
0.7720 (0.7720) Prec 75.781% (75.781%)

Epoch: [47][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.6174 (0.7467) Prec 79.688% (73.933%)

Epoch: [47][200/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
1.1523 (0.7494) Prec 59.375% (73.787%)

Epoch: [47][300/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss
0.6915 (0.7448) Prec 78.906% (73.876%)

Validation starts

Test: [0/79] Time 0.132 (0.132) Loss 0.7609 (0.7609) Prec 75.781%
(75.781%)

* Prec 71.850%

best acc: 73.950000

Epoch: [48][0/391] Time 0.209 (0.209) Data 0.172 (0.172) Loss
0.6729 (0.6729) Prec 79.688% (79.688%)

Epoch: [48][100/391] Time 0.053 (0.055) Data 0.001 (0.003) Loss
0.6139 (0.7512) Prec 82.031% (73.693%)

Epoch: [48][200/391] Time 0.053 (0.054) Data 0.002 (0.002) Loss
0.6465 (0.7532) Prec 76.562% (73.752%)

Epoch: [48][300/391] Time 0.053 (0.054) Data 0.001 (0.002) Loss
0.7516 (0.7501) Prec 75.000% (73.770%)

Validation starts

Test: [0/79] Time 0.114 (0.114) Loss 0.8141 (0.8141) Prec 72.656%
(72.656%)

* Prec 72.100%

best acc: 73.950000

Epoch: [49][0/391] Time 0.177 (0.177) Data 0.137 (0.137) Loss
0.6747 (0.6747) Prec 77.344% (77.344%)

Epoch: [49][100/391] Time 0.053 (0.055) Data 0.002 (0.003) Loss
0.6540 (0.7473) Prec 76.562% (74.080%)

Epoch: [49][200/391] Time 0.054 (0.054) Data 0.002 (0.002) Loss

```

0.6544 (0.7426)    Prec 78.125% (74.262%)
Epoch: [49][300/391]    Time 0.054 (0.054)    Data 0.002 (0.002)    Loss
0.7388 (0.7430)    Prec 74.219% (74.206%)
Validation starts
Test: [0/79]    Time 0.148 (0.148)    Loss 0.8164 (0.8164)    Prec 74.219%
(74.219%)
    * Prec 72.550%
best acc: 73.950000

```

```

[33]: model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 7255/10000 (73%)

```

[16]: PATH = "result/VGG16_quant/model_best.pth.tar"
checkpoint = torch.load(PATH, map_location="cuda")
model = VGG16_quant()
model.load_state_dict(checkpoint['state_dict'])
model.cuda()
model.eval()

structured_amount = 0.5
final_target_sparsity = 0.8

params_to_prune = []

for layer in model.modules():
    if isinstance(layer, QuantConv2d):
        prune.ln_structured(
            layer,
            name="weight",
            amount=structured_amount,

```

```

        n=1,
        dim=0,
    )
    w = layer.weight.detach()
    total = w.numel()
    zeros = (w == 0).sum().item()
    S_cur = zeros / total

    if S_cur >= final_target_sparsity:
        print(f"skip layer (already sparse): S_cur={S_cur:.3f}")
        continue

    S_t = final_target_sparsity
    extra = 1.0 - (1.0 - S_t) / (1.0 - S_cur)

    extra = max(0.0, min(1.0, extra))

    prune.l1_unstructured(
        layer,
        name="weight",
        amount=extra,
    )

```

```

[17]: def total_model_sparsity(model):
    total_elems = 0
    zero_elems = 0

    for m in model.modules():
        if isinstance(m, QuantConv2d):
            w = m.weight.data
            total_elems += w.numel()
            zero_elems += (w == 0).sum().item()

    return zero_elems / total_elems

print("Model Sparsity =", total_model_sparsity(model))

```

Model Sparsity = 0.7999999864042358

```

[18]: model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)

```

```

        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 1000/10000 (10%)

```

[19]: lr = 1e-1
weight_decay = 1e-4
epochs = 50
best_prec = 0

#model = nn.DataParallel(model).cuda()
model.cuda()
criterion = nn.CrossEntropyLoss().cuda()
optimizer = torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9,
    ↪weight_decay=weight_decay)
#cudnn.benchmark = True

if not os.path.exists('result'):
    os.makedirs('result')
fdir = 'result/'+'pruning'+str(model_name)
if not os.path.exists(fdir):
    os.makedirs(fdir)

for epoch in range(0, epochs):
    adjust_learning_rate(optimizer, epoch)

    train(trainloader, model, criterion, optimizer, epoch)

    # evaluate on test set
    print("Validation starts")
    prec = validate(testloader, model, criterion)

    # remember best precision and save checkpoint
    is_best = prec > best_prec
    best_prec = max(prec, best_prec)
    print('best acc: {:.1f}'.format(best_prec))
    save_checkpoint({
        'epoch': epoch + 1,

```

```

    'state_dict': model.state_dict(),
    'best_prec': best_prec,
    'optimizer': optimizer.state_dict(),
}, is_best, fdir)

```

```

Epoch: [0][0/391]      Time 0.187 (0.187)      Data 0.141 (0.141)      Loss
4.9563 (4.9563)      Prec 30.469% (30.469%)
Epoch: [0][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.6067 (1.0760)      Prec 81.250% (66.940%)
Epoch: [0][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.5991 (0.8075)      Prec 82.031% (74.343%)
Epoch: [0][300/391]    Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.3670 (0.6987)      Prec 89.062% (77.487%)
Validation starts
Test: [0/79]      Time 0.111 (0.111)      Loss 0.3827 (0.3827)      Prec 89.062%
(89.062%)
* Prec 83.300%
best acc: 83.300000
Epoch: [1][0/391]      Time 0.213 (0.213)      Data 0.157 (0.157)      Loss
0.2789 (0.2789)      Prec 86.719% (86.719%)
Epoch: [1][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.4926 (0.3826)      Prec 85.156% (87.283%)
Epoch: [1][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.3400 (0.3748)      Prec 87.500% (87.395%)
Epoch: [1][300/391]    Time 0.055 (0.056)      Data 0.002 (0.002)      Loss
0.3863 (0.3706)      Prec 88.281% (87.443%)
Validation starts
Test: [0/79]      Time 0.119 (0.119)      Loss 0.4381 (0.4381)      Prec 82.812%
(82.812%)
* Prec 84.540%
best acc: 84.540000
Epoch: [2][0/391]      Time 0.201 (0.201)      Data 0.150 (0.150)      Loss
0.2315 (0.2315)      Prec 92.188% (92.188%)
Epoch: [2][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.2489 (0.3308)      Prec 90.625% (88.420%)
Epoch: [2][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2099 (0.3296)      Prec 93.750% (88.546%)
Epoch: [2][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2546 (0.3228)      Prec 91.406% (88.728%)
Validation starts
Test: [0/79]      Time 0.137 (0.137)      Loss 0.5734 (0.5734)      Prec 82.812%
(82.812%)
* Prec 82.680%
best acc: 84.540000
Epoch: [3][0/391]      Time 0.221 (0.221)      Data 0.171 (0.171)      Loss
0.2458 (0.2458)      Prec 92.188% (92.188%)
Epoch: [3][100/391]    Time 0.055 (0.057)      Data 0.001 (0.003)      Loss

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0.2829 (0.3000) Prec 86.719% (89.921%)
Epoch: [3][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.4161 (0.3027) Prec 86.719% (89.758%)
Epoch: [3][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.2303 (0.3047) Prec 89.062% (89.646%)
Validation starts
Test: [0/79] Time 0.112 (0.112) Loss 0.3266 (0.3266) Prec 86.719%
(86.719%)
* Prec 84.300%
best acc: 84.540000
Epoch: [4][0/391] Time 0.185 (0.185) Data 0.144 (0.144) Loss
0.1528 (0.1528) Prec 95.312% (95.312%)
Epoch: [4][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.1428 (0.2757) Prec 96.094% (90.347%)
Epoch: [4][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2299 (0.2852) Prec 93.750% (90.054%)
Epoch: [4][300/391] Time 0.057 (0.056) Data 0.001 (0.002) Loss
0.2817 (0.2873) Prec 88.281% (90.054%)
Validation starts
Test: [0/79] Time 0.118 (0.118) Loss 0.4099 (0.4099) Prec 86.719%
(86.719%)
* Prec 84.730%
best acc: 84.730000
Epoch: [5][0/391] Time 0.170 (0.170) Data 0.130 (0.130) Loss
0.3069 (0.3069) Prec 89.844% (89.844%)
Epoch: [5][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.2625 (0.2629) Prec 90.625% (90.756%)
Epoch: [5][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.3071 (0.2713) Prec 89.062% (90.707%)
Epoch: [5][300/391] Time 0.057 (0.056) Data 0.002 (0.002) Loss
0.2523 (0.2734) Prec 89.062% (90.654%)
Validation starts
Test: [0/79] Time 0.136 (0.136) Loss 0.3524 (0.3524) Prec 89.062%
(89.062%)
* Prec 86.560%
best acc: 86.560000
Epoch: [6][0/391] Time 0.208 (0.208) Data 0.159 (0.159) Loss
0.3217 (0.3217) Prec 88.281% (88.281%)
Epoch: [6][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.2003 (0.2530) Prec 93.750% (91.081%)
Epoch: [6][200/391] Time 0.052 (0.056) Data 0.001 (0.002) Loss
0.1557 (0.2565) Prec 93.750% (91.103%)
Epoch: [6][300/391] Time 0.054 (0.056) Data 0.002 (0.002) Loss
0.2868 (0.2623) Prec 92.188% (90.895%)
Validation starts
Test: [0/79] Time 0.120 (0.120) Loss 0.4171 (0.4171) Prec 86.719%
(86.719%)
* Prec 85.550%

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best acc: 86.560000
Epoch: [7][0/391]      Time 0.197 (0.197)      Data 0.148 (0.148)      Loss
0.2629 (0.2629)      Prec 92.969% (92.969%)
Epoch: [7][100/391]    Time 0.052 (0.057)      Data 0.002 (0.003)      Loss
0.2936 (0.2490)      Prec 91.406% (91.592%)
Epoch: [7][200/391]    Time 0.058 (0.056)      Data 0.001 (0.002)      Loss
0.2228 (0.2513)      Prec 94.531% (91.379%)
Epoch: [7][300/391]    Time 0.058 (0.056)      Data 0.001 (0.002)      Loss
0.2311 (0.2535)      Prec 92.969% (91.339%)
Validation starts
Test: [0/79]      Time 0.125 (0.125)      Loss 0.4229 (0.4229)      Prec 88.281%
(88.281%)
* Prec 86.000%
best acc: 86.560000
Epoch: [8][0/391]      Time 0.185 (0.185)      Data 0.145 (0.145)      Loss
0.1675 (0.1675)      Prec 96.094% (96.094%)
Epoch: [8][100/391]    Time 0.055 (0.057)      Data 0.001 (0.003)      Loss
0.2481 (0.2343)      Prec 91.406% (91.979%)
Epoch: [8][200/391]    Time 0.055 (0.056)      Data 0.002 (0.002)      Loss
0.2208 (0.2345)      Prec 91.406% (91.849%)
Epoch: [8][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1764 (0.2387)      Prec 92.188% (91.829%)
Validation starts
Test: [0/79]      Time 0.113 (0.113)      Loss 0.3208 (0.3208)      Prec 87.500%
(87.500%)
* Prec 86.350%
best acc: 86.560000
Epoch: [9][0/391]      Time 0.181 (0.181)      Data 0.138 (0.138)      Loss
0.2067 (0.2067)      Prec 94.531% (94.531%)
Epoch: [9][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.2722 (0.2310)      Prec 91.406% (92.280%)
Epoch: [9][200/391]    Time 0.063 (0.056)      Data 0.001 (0.002)      Loss
0.1996 (0.2403)      Prec 92.188% (91.772%)
Epoch: [9][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2480 (0.2418)      Prec 92.188% (91.780%)
Validation starts
Test: [0/79]      Time 0.117 (0.117)      Loss 0.3264 (0.3264)      Prec 89.062%
(89.062%)
* Prec 85.740%
best acc: 86.560000
Epoch: [10][0/391]     Time 0.196 (0.196)      Data 0.152 (0.152)      Loss
0.1767 (0.1767)      Prec 92.969% (92.969%)
Epoch: [10][100/391]   Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.2601 (0.2418)      Prec 89.062% (91.545%)
Epoch: [10][200/391]   Time 0.057 (0.056)      Data 0.001 (0.002)      Loss
0.1561 (0.2414)      Prec 96.094% (91.535%)
Epoch: [10][300/391]   Time 0.055 (0.056)      Data 0.001 (0.002)      Loss
0.2719 (0.2436)      Prec 89.062% (91.497%)

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Validation starts

Test: [0/79] Time 0.122 (0.122) Loss 0.4207 (0.4207) Prec 85.156%
(85.156%)

* Prec 84.720%

best acc: 86.560000

Epoch: [11][0/391]	Time 0.185 (0.185)	Data 0.148 (0.148)	Loss
0.3000 (0.3000)	Prec 89.844% (89.844%)		
Epoch: [11][100/391]	Time 0.055 (0.056)	Data 0.001 (0.003)	Loss
0.3041 (0.2335)	Prec 89.062% (91.754%)		
Epoch: [11][200/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.1192 (0.2355)	Prec 96.094% (91.861%)		
Epoch: [11][300/391]	Time 0.055 (0.056)	Data 0.002 (0.002)	Loss
0.2696 (0.2391)	Prec 88.281% (91.746%)		

Validation starts

Test: [0/79] Time 0.119 (0.119) Loss 0.4088 (0.4088) Prec 89.062%
(89.062%)

* Prec 86.640%

best acc: 86.640000

Epoch: [12][0/391]	Time 0.189 (0.189)	Data 0.145 (0.145)	Loss
0.1658 (0.1658)	Prec 95.312% (95.312%)		
Epoch: [12][100/391]	Time 0.054 (0.057)	Data 0.001 (0.003)	Loss
0.2275 (0.2313)	Prec 92.188% (91.824%)		
Epoch: [12][200/391]	Time 0.053 (0.056)	Data 0.001 (0.002)	Loss
0.2378 (0.2311)	Prec 89.844% (91.927%)		
Epoch: [12][300/391]	Time 0.053 (0.056)	Data 0.001 (0.002)	Loss
0.1643 (0.2334)	Prec 96.094% (91.840%)		

Validation starts

Test: [0/79] Time 0.117 (0.117) Loss 0.3955 (0.3955) Prec 86.719%
(86.719%)

* Prec 85.700%

best acc: 86.640000

Epoch: [13][0/391]	Time 0.177 (0.177)	Data 0.137 (0.137)	Loss
0.1550 (0.1550)	Prec 94.531% (94.531%)		
Epoch: [13][100/391]	Time 0.055 (0.057)	Data 0.001 (0.003)	Loss
0.2089 (0.2193)	Prec 90.625% (92.443%)		
Epoch: [13][200/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.2836 (0.2278)	Prec 90.625% (92.102%)		
Epoch: [13][300/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.2512 (0.2297)	Prec 92.188% (91.980%)		

Validation starts

Test: [0/79] Time 0.113 (0.113) Loss 0.3788 (0.3788) Prec 86.719%
(86.719%)

* Prec 86.250%

best acc: 86.640000

Epoch: [14][0/391]	Time 0.162 (0.162)	Data 0.122 (0.122)	Loss
0.2845 (0.2845)	Prec 91.406% (91.406%)		
Epoch: [14][100/391]	Time 0.055 (0.057)	Data 0.002 (0.003)	Loss
0.2352 (0.2171)	Prec 92.188% (92.435%)		

Epoch: [14][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.2240 (0.2250) Prec 91.406% (92.129%)

Epoch: [14][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1854 (0.2263) Prec 92.188% (92.089%)

Validation starts

Test: [0/79] Time 0.121 (0.121) Loss 0.3903 (0.3903) Prec 86.719%
(86.719%)

* Prec 85.560%

best acc: 86.640000

Epoch: [15][0/391] Time 0.177 (0.177) Data 0.129 (0.129) Loss
0.2041 (0.2041) Prec 94.531% (94.531%)

Epoch: [15][100/391] Time 0.057 (0.057) Data 0.002 (0.003) Loss
0.1450 (0.2174) Prec 96.094% (92.273%)

Epoch: [15][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.3190 (0.2184) Prec 89.844% (92.195%)

Epoch: [15][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.2359 (0.2236) Prec 90.625% (92.086%)

Validation starts

Test: [0/79] Time 0.120 (0.120) Loss 0.3211 (0.3211) Prec 89.844%
(89.844%)

* Prec 87.160%

best acc: 87.160000

Epoch: [16][0/391] Time 0.176 (0.176) Data 0.132 (0.132) Loss
0.2335 (0.2335) Prec 91.406% (91.406%)

Epoch: [16][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.3114 (0.2104) Prec 90.625% (92.474%)

Epoch: [16][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1682 (0.2144) Prec 93.750% (92.432%)

Epoch: [16][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.2507 (0.2182) Prec 92.188% (92.325%)

Validation starts

Test: [0/79] Time 0.119 (0.119) Loss 0.3076 (0.3076) Prec 89.062%
(89.062%)

* Prec 86.340%

best acc: 87.160000

Epoch: [17][0/391] Time 0.199 (0.199) Data 0.157 (0.157) Loss
0.2313 (0.2313) Prec 94.531% (94.531%)

Epoch: [17][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.2443 (0.2033) Prec 91.406% (93.015%)

Epoch: [17][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.2552 (0.2114) Prec 91.406% (92.763%)

Epoch: [17][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1939 (0.2167) Prec 92.969% (92.517%)

Validation starts

Test: [0/79] Time 0.121 (0.121) Loss 0.4078 (0.4078) Prec 88.281%
(88.281%)

* Prec 86.490%

best acc: 87.160000

Epoch: [18][0/391] Time 0.192 (0.192) Data 0.154 (0.154) Loss
 0.2002 (0.2002) Prec 92.969% (92.969%)
 Epoch: [18][100/391] Time 0.058 (0.057) Data 0.001 (0.003) Loss
 0.3016 (0.2063) Prec 89.062% (92.837%)
 Epoch: [18][200/391] Time 0.057 (0.056) Data 0.001 (0.002) Loss
 0.2929 (0.2094) Prec 90.625% (92.802%)
 Epoch: [18][300/391] Time 0.058 (0.056) Data 0.001 (0.002) Loss
 0.1706 (0.2140) Prec 93.750% (92.546%)
 Validation starts
 Test: [0/79] Time 0.118 (0.118) Loss 0.5444 (0.5444) Prec 81.250%
 (81.250%)
 * Prec 85.680%
 best acc: 87.160000
 Epoch: [19][0/391] Time 0.177 (0.177) Data 0.138 (0.138) Loss
 0.2825 (0.2825) Prec 92.969% (92.969%)
 Epoch: [19][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.2831 (0.2016) Prec 91.406% (93.007%)
 Epoch: [19][200/391] Time 0.055 (0.057) Data 0.002 (0.002) Loss
 0.2524 (0.2076) Prec 90.625% (92.701%)
 Epoch: [19][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
 0.1945 (0.2134) Prec 94.531% (92.553%)
 Validation starts
 Test: [0/79] Time 0.119 (0.119) Loss 0.3368 (0.3368) Prec 89.844%
 (89.844%)
 * Prec 87.050%
 best acc: 87.160000
 Epoch: [20][0/391] Time 0.201 (0.201) Data 0.152 (0.152) Loss
 0.2304 (0.2304) Prec 90.625% (90.625%)
 Epoch: [20][100/391] Time 0.060 (0.057) Data 0.001 (0.003) Loss
 0.2472 (0.2089) Prec 92.969% (92.659%)
 Epoch: [20][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
 0.2446 (0.2090) Prec 89.062% (92.708%)
 Epoch: [20][300/391] Time 0.051 (0.056) Data 0.001 (0.002) Loss
 0.3094 (0.2160) Prec 89.844% (92.457%)
 Validation starts
 Test: [0/79] Time 0.114 (0.114) Loss 0.3544 (0.3544) Prec 88.281%
 (88.281%)
 * Prec 86.910%
 best acc: 87.160000
 Epoch: [21][0/391] Time 0.186 (0.186) Data 0.137 (0.137) Loss
 0.1669 (0.1669) Prec 93.750% (93.750%)
 Epoch: [21][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.2637 (0.2010) Prec 88.281% (93.108%)
 Epoch: [21][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.1157 (0.2069) Prec 95.312% (92.864%)
 Epoch: [21][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.2141 (0.2115) Prec 93.750% (92.761%)
 Validation starts

Test: [0/79] Time 0.148 (0.148) Loss 0.2376 (0.2376) Prec 92.188%
(92.188%)

* Prec 87.340%

best acc: 87.340000

Epoch: [22][0/391] Time 0.204 (0.204) Data 0.156 (0.156) Loss
0.1540 (0.1540) Prec 95.312% (95.312%)

Epoch: [22][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
0.1233 (0.2130) Prec 96.094% (92.597%)

Epoch: [22][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.2036 (0.2102) Prec 93.750% (92.681%)

Epoch: [22][300/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1918 (0.2130) Prec 92.969% (92.540%)

Validation starts

Test: [0/79] Time 0.112 (0.112) Loss 0.3638 (0.3638) Prec 90.625%
(90.625%)

* Prec 87.200%

best acc: 87.340000

Epoch: [23][0/391] Time 0.214 (0.214) Data 0.165 (0.165) Loss
0.2597 (0.2597) Prec 91.406% (91.406%)

Epoch: [23][100/391] Time 0.056 (0.057) Data 0.001 (0.003) Loss
0.1830 (0.2034) Prec 95.312% (92.721%)

Epoch: [23][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.1120 (0.2075) Prec 96.094% (92.600%)

Epoch: [23][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2167 (0.2148) Prec 92.188% (92.416%)

Validation starts

Test: [0/79] Time 0.115 (0.115) Loss 0.3154 (0.3154) Prec 90.625%
(90.625%)

* Prec 86.830%

best acc: 87.340000

Epoch: [24][0/391] Time 0.203 (0.203) Data 0.149 (0.149) Loss
0.4408 (0.4408) Prec 87.500% (87.500%)

Epoch: [24][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.1368 (0.2013) Prec 95.312% (92.899%)

Epoch: [24][200/391] Time 0.056 (0.056) Data 0.001 (0.002) Loss
0.1156 (0.2047) Prec 96.875% (92.840%)

Epoch: [24][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1880 (0.2067) Prec 93.750% (92.777%)

Validation starts

Test: [0/79] Time 0.121 (0.121) Loss 0.3327 (0.3327) Prec 93.750%
(93.750%)

* Prec 86.980%

best acc: 87.340000

Epoch: [25][0/391] Time 0.206 (0.206) Data 0.154 (0.154) Loss
0.1792 (0.1792) Prec 93.750% (93.750%)

Epoch: [25][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.2411 (0.1971) Prec 92.188% (93.077%)

Epoch: [25][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss

0.2165 (0.2060) Prec 91.406% (92.755%)
Epoch: [25][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2161 (0.2118) Prec 92.969% (92.515%)
Validation starts
Test: [0/79] Time 0.119 (0.119) Loss 0.3948 (0.3948) Prec 88.281%
(88.281%)
* Prec 87.730%
best acc: 87.730000
Epoch: [26][0/391] Time 0.189 (0.189) Data 0.148 (0.148) Loss
0.1399 (0.1399) Prec 95.312% (95.312%)
Epoch: [26][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.2981 (0.1985) Prec 90.625% (93.239%)
Epoch: [26][200/391] Time 0.054 (0.057) Data 0.001 (0.002) Loss
0.2292 (0.2014) Prec 92.969% (93.120%)
Epoch: [26][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1230 (0.2074) Prec 96.875% (92.847%)
Validation starts
Test: [0/79] Time 0.119 (0.119) Loss 0.4049 (0.4049) Prec 86.719%
(86.719%)
* Prec 87.070%
best acc: 87.730000
Epoch: [27][0/391] Time 0.163 (0.163) Data 0.124 (0.124) Loss
0.1553 (0.1553) Prec 92.969% (92.969%)
Epoch: [27][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.1800 (0.1941) Prec 96.094% (93.023%)
Epoch: [27][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2400 (0.1999) Prec 90.625% (92.864%)
Epoch: [27][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1747 (0.2037) Prec 94.531% (92.834%)
Validation starts
Test: [0/79] Time 0.113 (0.113) Loss 0.3711 (0.3711) Prec 87.500%
(87.500%)
* Prec 87.180%
best acc: 87.730000
Epoch: [28][0/391] Time 0.190 (0.190) Data 0.141 (0.141) Loss
0.1798 (0.1798) Prec 92.188% (92.188%)
Epoch: [28][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
0.2767 (0.1929) Prec 92.188% (93.340%)
Epoch: [28][200/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.2196 (0.2020) Prec 92.969% (93.132%)
Epoch: [28][300/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
0.2309 (0.2057) Prec 91.406% (92.878%)
Validation starts
Test: [0/79] Time 0.115 (0.115) Loss 0.2671 (0.2671) Prec 91.406%
(91.406%)
* Prec 87.690%
best acc: 87.730000
Epoch: [29][0/391] Time 0.213 (0.213) Data 0.160 (0.160) Loss

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0.2245 (0.2245)    Prec 92.969% (92.969%)
Epoch: [29][100/391]    Time 0.055 (0.057)    Data 0.001 (0.003)    Loss
0.1361 (0.2033)    Prec 94.531% (92.899%)
Epoch: [29][200/391]    Time 0.055 (0.056)    Data 0.002 (0.002)    Loss
0.1352 (0.2028)    Prec 95.312% (92.922%)
Epoch: [29][300/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.2684 (0.2081)    Prec 92.969% (92.792%)
Validation starts
Test: [0/79]    Time 0.146 (0.146)    Loss 0.3721 (0.3721)    Prec 89.062%
(89.062%)
* Prec 87.300%
best acc: 87.730000
Epoch: [30][0/391]    Time 0.212 (0.212)    Data 0.171 (0.171)    Loss
0.1705 (0.1705)    Prec 94.531% (94.531%)
Epoch: [30][100/391]    Time 0.055 (0.057)    Data 0.001 (0.003)    Loss
0.1892 (0.1948)    Prec 94.531% (93.325%)
Epoch: [30][200/391]    Time 0.055 (0.056)    Data 0.001 (0.002)    Loss
0.2632 (0.2011)    Prec 90.625% (93.078%)
Epoch: [30][300/391]    Time 0.056 (0.056)    Data 0.001 (0.002)    Loss
0.1758 (0.2016)    Prec 93.750% (92.990%)
Validation starts
Test: [0/79]    Time 0.117 (0.117)    Loss 0.3504 (0.3504)    Prec 87.500%
(87.500%)
* Prec 86.880%
best acc: 87.730000
Epoch: [31][0/391]    Time 0.221 (0.221)    Data 0.174 (0.174)    Loss
0.1546 (0.1546)    Prec 96.094% (96.094%)
Epoch: [31][100/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1038 (0.2108)    Prec 95.312% (92.567%)
Epoch: [31][200/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.2735 (0.2124)    Prec 90.625% (92.533%)
Epoch: [31][300/391]    Time 0.056 (0.056)    Data 0.002 (0.002)    Loss
0.2815 (0.2103)    Prec 92.188% (92.634%)
Validation starts
Test: [0/79]    Time 0.137 (0.137)    Loss 0.3773 (0.3773)    Prec 89.062%
(89.062%)
* Prec 86.150%
best acc: 87.730000
Epoch: [32][0/391]    Time 0.214 (0.214)    Data 0.161 (0.161)    Loss
0.2014 (0.2014)    Prec 92.969% (92.969%)
Epoch: [32][100/391]    Time 0.056 (0.058)    Data 0.002 (0.003)    Loss
0.2019 (0.1986)    Prec 91.406% (93.201%)
Epoch: [32][200/391]    Time 0.056 (0.057)    Data 0.002 (0.003)    Loss
0.1968 (0.2026)    Prec 89.844% (92.895%)
Epoch: [32][300/391]    Time 0.056 (0.056)    Data 0.002 (0.002)    Loss
0.2261 (0.2051)    Prec 91.406% (92.865%)
Validation starts
Test: [0/79]    Time 0.127 (0.127)    Loss 0.3542 (0.3542)    Prec 87.500%

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(87.500%)

* Prec 87.220%

best acc: 87.730000

Epoch: [33][0/391]	Time 0.208 (0.208)	Data 0.158 (0.158)	Loss
0.1921 (0.1921)	Prec 91.406% (91.406%)		
Epoch: [33][100/391]	Time 0.055 (0.057)	Data 0.001 (0.003)	Loss
0.2427 (0.1973)	Prec 93.750% (92.891%)		
Epoch: [33][200/391]	Time 0.056 (0.056)	Data 0.001 (0.002)	Loss
0.1411 (0.1938)	Prec 96.094% (93.105%)		
Epoch: [33][300/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.2635 (0.1979)	Prec 89.844% (92.995%)		

Validation starts

Test: [0/79] Time 0.115 (0.115) Loss 0.2786 (0.2786) Prec 91.406% (91.406%)

* Prec 87.530%

best acc: 87.730000

Epoch: [34][0/391]	Time 0.196 (0.196)	Data 0.149 (0.149)	Loss
0.1805 (0.1805)	Prec 93.750% (93.750%)		
Epoch: [34][100/391]	Time 0.053 (0.057)	Data 0.001 (0.003)	Loss
0.1152 (0.1925)	Prec 95.312% (93.062%)		
Epoch: [34][200/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.2738 (0.1971)	Prec 88.281% (93.015%)		
Epoch: [34][300/391]	Time 0.055 (0.056)	Data 0.001 (0.002)	Loss
0.1685 (0.1993)	Prec 95.312% (93.018%)		

Validation starts

Test: [0/79] Time 0.118 (0.118) Loss 0.3636 (0.3636) Prec 86.719% (86.719%)

* Prec 86.420%

best acc: 87.730000

Epoch: [35][0/391]	Time 0.214 (0.214)	Data 0.172 (0.172)	Loss
0.0806 (0.0806)	Prec 96.875% (96.875%)		
Epoch: [35][100/391]	Time 0.055 (0.058)	Data 0.001 (0.003)	Loss
0.2156 (0.2063)	Prec 92.969% (92.768%)		
Epoch: [35][200/391]	Time 0.055 (0.057)	Data 0.002 (0.003)	Loss
0.1603 (0.2037)	Prec 93.750% (92.786%)		
Epoch: [35][300/391]	Time 0.051 (0.056)	Data 0.002 (0.002)	Loss
0.2747 (0.2074)	Prec 89.844% (92.748%)		

Validation starts

Test: [0/79] Time 0.128 (0.128) Loss 0.4195 (0.4195) Prec 85.156% (85.156%)

* Prec 85.570%

best acc: 87.730000

Epoch: [36][0/391]	Time 0.179 (0.179)	Data 0.134 (0.134)	Loss
0.2276 (0.2276)	Prec 92.969% (92.969%)		
Epoch: [36][100/391]	Time 0.055 (0.057)	Data 0.001 (0.003)	Loss
0.1648 (0.2025)	Prec 92.969% (92.992%)		
Epoch: [36][200/391]	Time 0.056 (0.056)	Data 0.002 (0.002)	Loss
0.4078 (0.2022)	Prec 85.938% (92.910%)		

Epoch: [36][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.1490 (0.2029) Prec 93.750% (92.810%)
Validation starts
Test: [0/79] Time 0.110 (0.110) Loss 0.3616 (0.3616) Prec 89.844%
(89.844%)
* Prec 87.460%
best acc: 87.730000
Epoch: [37][0/391] Time 0.181 (0.181) Data 0.141 (0.141) Loss
0.2115 (0.2115) Prec 92.969% (92.969%)
Epoch: [37][100/391] Time 0.057 (0.057) Data 0.002 (0.003) Loss
0.2121 (0.1907) Prec 91.406% (93.123%)
Epoch: [37][200/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1429 (0.1905) Prec 95.312% (93.287%)
Epoch: [37][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
0.1880 (0.1975) Prec 92.969% (93.075%)
Validation starts
Test: [0/79] Time 0.132 (0.132) Loss 0.3253 (0.3253) Prec 89.062%
(89.062%)
* Prec 87.090%
best acc: 87.730000
Epoch: [38][0/391] Time 0.187 (0.187) Data 0.143 (0.143) Loss
0.2398 (0.2398) Prec 89.062% (89.062%)
Epoch: [38][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
0.2546 (0.1928) Prec 89.062% (93.255%)
Epoch: [38][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
0.2550 (0.1993) Prec 92.188% (93.023%)
Epoch: [38][300/391] Time 0.058 (0.056) Data 0.002 (0.002) Loss
0.1460 (0.2026) Prec 96.094% (92.873%)
Validation starts
Test: [0/79] Time 0.120 (0.120) Loss 0.2877 (0.2877) Prec 89.062%
(89.062%)
* Prec 88.440%
best acc: 88.440000
Epoch: [39][0/391] Time 0.211 (0.211) Data 0.162 (0.162) Loss
0.2411 (0.2411) Prec 91.406% (91.406%)
Epoch: [39][100/391] Time 0.054 (0.057) Data 0.001 (0.003) Loss
0.1807 (0.1823) Prec 94.531% (93.704%)
Epoch: [39][200/391] Time 0.054 (0.056) Data 0.001 (0.002) Loss
0.2030 (0.1936) Prec 92.188% (93.260%)
Epoch: [39][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
0.1442 (0.1982) Prec 96.094% (93.096%)
Validation starts
Test: [0/79] Time 0.115 (0.115) Loss 0.3809 (0.3809) Prec 88.281%
(88.281%)
* Prec 85.840%
best acc: 88.440000
Epoch: [40][0/391] Time 0.190 (0.190) Data 0.150 (0.150) Loss
0.1719 (0.1719) Prec 91.406% (91.406%)

Epoch: [40][100/391] Time 0.055 (0.057) Data 0.002 (0.003) Loss
 0.1877 (0.1871) Prec 92.969% (93.526%)
 Epoch: [40][200/391] Time 0.056 (0.057) Data 0.002 (0.002) Loss
 0.1559 (0.1911) Prec 93.750% (93.501%)
 Epoch: [40][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.1966 (0.1975) Prec 92.188% (93.174%)
 Validation starts
 Test: [0/79] Time 0.127 (0.127) Loss 0.4541 (0.4541) Prec 85.938%
 (85.938%)
 * Prec 86.550%
 best acc: 88.440000
 Epoch: [41][0/391] Time 0.191 (0.191) Data 0.144 (0.144) Loss
 0.2275 (0.2275) Prec 92.969% (92.969%)
 Epoch: [41][100/391] Time 0.055 (0.057) Data 0.001 (0.003) Loss
 0.2111 (0.2030) Prec 93.750% (92.915%)
 Epoch: [41][200/391] Time 0.055 (0.056) Data 0.001 (0.002) Loss
 0.3408 (0.1985) Prec 89.844% (92.953%)
 Epoch: [41][300/391] Time 0.056 (0.056) Data 0.002 (0.002) Loss
 0.1514 (0.2001) Prec 93.750% (92.925%)
 Validation starts
 Test: [0/79] Time 0.138 (0.138) Loss 0.4232 (0.4232) Prec 89.062%
 (89.062%)
 * Prec 85.330%
 best acc: 88.440000
 Epoch: [42][0/391] Time 0.197 (0.197) Data 0.149 (0.149) Loss
 0.2037 (0.2037) Prec 92.188% (92.188%)
 Epoch: [42][100/391] Time 0.058 (0.057) Data 0.001 (0.003) Loss
 0.2320 (0.1962) Prec 90.625% (93.162%)
 Epoch: [42][200/391] Time 0.057 (0.056) Data 0.001 (0.002) Loss
 0.2115 (0.1994) Prec 93.750% (93.027%)
 Epoch: [42][300/391] Time 0.058 (0.056) Data 0.001 (0.002) Loss
 0.1980 (0.2019) Prec 89.844% (92.816%)
 Validation starts
 Test: [0/79] Time 0.122 (0.122) Loss 0.3096 (0.3096) Prec 92.188%
 (92.188%)
 * Prec 87.670%
 best acc: 88.440000
 Epoch: [43][0/391] Time 0.168 (0.168) Data 0.127 (0.127) Loss
 0.1701 (0.1701) Prec 92.969% (92.969%)
 Epoch: [43][100/391] Time 0.056 (0.057) Data 0.002 (0.003) Loss
 0.1888 (0.1909) Prec 91.406% (93.270%)
 Epoch: [43][200/391] Time 0.059 (0.056) Data 0.001 (0.002) Loss
 0.2046 (0.1927) Prec 93.750% (93.256%)
 Epoch: [43][300/391] Time 0.055 (0.056) Data 0.002 (0.002) Loss
 0.2463 (0.1998) Prec 89.844% (92.979%)
 Validation starts
 Test: [0/79] Time 0.122 (0.122) Loss 0.3275 (0.3275) Prec 89.062%
 (89.062%)


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* Prec 86.520%
best acc: 88.440000
Epoch: [44][0/391]      Time 0.198 (0.198)      Data 0.147 (0.147)      Loss
0.1425 (0.1425)      Prec 93.750% (93.750%)
Epoch: [44][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.2083 (0.1835)      Prec 93.750% (93.363%)
Epoch: [44][200/391]    Time 0.056 (0.057)      Data 0.002 (0.002)      Loss
0.2245 (0.1949)      Prec 90.625% (93.046%)
Epoch: [44][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1839 (0.2000)      Prec 95.312% (93.002%)
Validation starts
Test: [0/79]      Time 0.119 (0.119)      Loss 0.4191 (0.4191)      Prec 85.156%
(85.156%)
* Prec 86.360%
best acc: 88.440000
Epoch: [45][0/391]      Time 0.233 (0.233)      Data 0.185 (0.185)      Loss
0.2673 (0.2673)      Prec 90.625% (90.625%)
Epoch: [45][100/391]    Time 0.056 (0.057)      Data 0.002 (0.003)      Loss
0.1127 (0.1871)      Prec 96.875% (93.487%)
Epoch: [45][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.1429 (0.1972)      Prec 93.750% (93.132%)
Epoch: [45][300/391]    Time 0.059 (0.056)      Data 0.001 (0.002)      Loss
0.1827 (0.2011)      Prec 92.969% (92.909%)
Validation starts
Test: [0/79]      Time 0.126 (0.126)      Loss 0.2907 (0.2907)      Prec 89.844%
(89.844%)
* Prec 86.240%
best acc: 88.440000
Epoch: [46][0/391]      Time 0.194 (0.194)      Data 0.152 (0.152)      Loss
0.1900 (0.1900)      Prec 92.969% (92.969%)
Epoch: [46][100/391]    Time 0.055 (0.057)      Data 0.002 (0.003)      Loss
0.2026 (0.1844)      Prec 94.531% (93.673%)
Epoch: [46][200/391]    Time 0.055 (0.057)      Data 0.002 (0.002)      Loss
0.2200 (0.1931)      Prec 89.844% (93.245%)
Epoch: [46][300/391]    Time 0.049 (0.056)      Data 0.002 (0.002)      Loss
0.3069 (0.1953)      Prec 89.062% (93.080%)
Validation starts
Test: [0/79]      Time 0.111 (0.111)      Loss 0.3956 (0.3956)      Prec 89.062%
(89.062%)
* Prec 87.740%
best acc: 88.440000
Epoch: [47][0/391]      Time 0.187 (0.187)      Data 0.139 (0.139)      Loss
0.1776 (0.1776)      Prec 92.188% (92.188%)
Epoch: [47][100/391]    Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.1875 (0.1933)      Prec 95.312% (93.410%)
Epoch: [47][200/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2222 (0.1934)      Prec 91.406% (93.198%)
Epoch: [47][300/391]    Time 0.056 (0.056)      Data 0.001 (0.002)      Loss

```

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0.2253 (0.1962)    Prec 91.406% (93.080%)
Validation starts
Test: [0/79]      Time 0.122 (0.122)      Loss 0.4029 (0.4029)    Prec 89.062%
(89.062%)
* Prec 87.140%
best acc: 88.440000
Epoch: [48][0/391]    Time 0.157 (0.157)      Data 0.120 (0.120)      Loss
0.1769 (0.1769)    Prec 93.750% (93.750%)
Epoch: [48][100/391]  Time 0.055 (0.056)      Data 0.002 (0.003)      Loss
0.1882 (0.1930)    Prec 92.188% (93.131%)
Epoch: [48][200/391]  Time 0.056 (0.056)      Data 0.001 (0.002)      Loss
0.2080 (0.1943)    Prec 93.750% (93.210%)
Epoch: [48][300/391]  Time 0.055 (0.056)      Data 0.002 (0.002)      Loss
0.1980 (0.1995)    Prec 92.188% (92.992%)
Validation starts
Test: [0/79]      Time 0.119 (0.119)      Loss 0.3297 (0.3297)    Prec 89.844%
(89.844%)
* Prec 87.100%
best acc: 88.440000
Epoch: [49][0/391]    Time 0.211 (0.211)      Data 0.155 (0.155)      Loss
0.1440 (0.1440)    Prec 96.094% (96.094%)
Epoch: [49][100/391]  Time 0.056 (0.057)      Data 0.001 (0.003)      Loss
0.2361 (0.2008)    Prec 90.625% (92.884%)
Epoch: [49][200/391]  Time 0.055 (0.057)      Data 0.002 (0.002)      Loss
0.0968 (0.1975)    Prec 97.656% (92.977%)
Epoch: [49][300/391]  Time 0.056 (0.056)      Data 0.002 (0.002)      Loss
0.1550 (0.2001)    Prec 96.094% (92.961%)
Validation starts
Test: [0/79]      Time 0.114 (0.114)      Loss 0.3652 (0.3652)    Prec 89.844%
(89.844%)
* Prec 87.240%
best acc: 88.440000

```

```

[20]: model.eval()
test_loss = 0
correct = 0
with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),

```

```
100. * correct / len(testloader.dataset)))
```

Test set: Accuracy: 8724/10000 (87%)

```
[21]: ### Check sparsity ###  
mask1 = model.features[40].weight_mask  
sparsity_mask1 = (mask1 == 0).sum() / mask1.nelement()  
  
print("Sparsity level: ", sparsity_mask1)
```

Sparsity level: tensor(0.8000, device='cuda:0')

[]:

[]: